

GUIDANCE COUNSELING MANAGEMENT SYSTEM FOR CATSU

A Capstone Project
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CHAPTER 1

INTRODUCTION

Project Context

Guidance counseling is a vital process that helps students understand themselves, make informed decisions, and develop the skills necessary to handle personal, academic, and social challenges. It supports students' holistic development by guiding them in achieving their educational and personal goals. According to Alegado and Alcantara (2021), guidance counseling contributes significantly to students' success by enhancing their self-awareness, personality, and emotional well-being.

At Catanduanes State University (CatSU), the Guidance and Counseling Office (GCTO) continue to provide services that help students cope with academic pressure, career uncertainties, and personal concerns. However, the current **walk-in appointment system** often leads to scheduling conflicts, manual record-keeping errors, and longer processing times. These inefficiencies can delay the delivery of timely support and increase administrative workload for guidance staff.

To address these challenges, this study proposes the development of a **Guidance Counseling Management System (GCMS)** — a web-based platform designed to **automate scheduling, digitalize case documentation, and generate counseling reports**. The system aims to minimize paperwork, reduce scheduling conflicts, and enhance

accessibility to counseling services. As highlighted by Abiog et al. (2024), adopting digital systems in guidance offices can significantly improve efficiency, data management, and service delivery.

Through the GCMS, students will be able to easily book appointments online, while counselors can manage records and monitor student progress more effectively. This initiative supports CatSU's goal of providing student-centered, technology-driven services that promote academic success and overall well-being.

Purpose and Description

The **Guidance and Counseling Office (GCTO)** of Catanduanes State University (CatSU) currently relies on a manual, walk-in process for managing student counseling services. This traditional system often leads to scheduling conflicts, lost records, and delays in assisting students who require immediate guidance. The manual handling of appointment forms and counseling data also increases the workload of guidance staff, making it difficult to efficiently track student progress and maintain organized records.

To address these limitations, this study aims to develop a **web-based Guidance Counseling Management System (GCMS)** that automates the scheduling and record-keeping processes. The system integrates core features such as online appointment booking, digital case management, and automated report generation. By transitioning to a

digital platform, the GCTO can streamline its workflow, ensure data accuracy, and provide a more organized and efficient service delivery for both students and counselors.

Furthermore, the system seeks to enhance accessibility and transparency in guidance operations. Students will be able to schedule counseling sessions conveniently online, while counselors can monitor and update case files in real time. The system's data-driven reporting capability will also support administrators in evaluating counseling trends and improving service outcomes. Overall, the GCMS aligns with CatSU's commitment to delivering innovative, student-centered services that promote academic success and personal development.

Objectives

The primary objective of this project is to design and implement an automated **Guidance Counseling Management System (GCMS)** that simplifies the process of booking and managing counseling sessions.

Specifically, this study aims to:

1. Develop an intuitive and efficient web-based scheduling system that automates appointment booking for students and counselors.
2. Implement digital case management for accurate record-keeping and monitoring of counseling sessions.

3. Generate reports summarizing counseling activities to support decision-making and service evaluation.
4. Evaluate the system's quality and effectiveness using the **ISO/IEC 25010 software quality model**, focusing on functionality, usability, performance efficiency, reliability, and security.

Scope and Limitations

The scope of this study covers the design, development, and implementation of a web-based Guidance Counseling Management System for CatSU. The system's core modules include appointment scheduling, counselor case management, and report generation. It caters to three user roles: students, counselors, and administrators. The GCMS provides online booking features, automated notifications, and centralized data storage to enhance accessibility, efficiency, and service monitoring.

The study is limited to the system's technical and operational aspects within CatSU. It does not measure the psychological effectiveness of counseling sessions but focuses on the functionality and usability of the platform. The system's deployment is site-specific, meaning findings may not fully apply to other institutions. Furthermore, its performance depends on the university's IT infrastructure and the users' adaptability to digital systems.

Definition of Terms

This section defines key terms used in the study to ensure clarity and consistency in interpreting the concepts and processes involved in the Guidance Counseling Management System.

Case Management – A system feature that allows counselors to document, organize, and monitor students' counseling histories and progress.

Compatibility – The ability of the system to function effectively with other software, hardware, or environments.

Functional Suitability – The extent to which the system provides functions that meet the specified user needs.

Usability – The ease with which users can learn, navigate, and interact with the system.

Performance Efficiency – The system's ability to perform tasks quickly and effectively while utilizing resources optimally.

Reliability – The consistency of the system's performance over time without failures.

Security – The capability of the system to protect information and prevent unauthorized access.

Maintainability – The ease of modifying or updating the system to correct issues or enhance performance.

Portability – The ability of the system to be deployed or adapted across different environments or platforms.

ISO/IEC 25010 – An international standard for evaluating software quality through eight key characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Conceptual Framework

The conceptual framework serves as the foundation for understanding how the **Guidance Counseling Management System (GCMS)** operates and achieves its objectives. It illustrates the logical relationship between the key components of the study from the resources needed for development to the processes involved and the expected outcomes. This framework provides a structured overview of how the system is designed, developed, and evaluated to improve the efficiency of guidance counseling services at Catanduanes State University (CatSU).

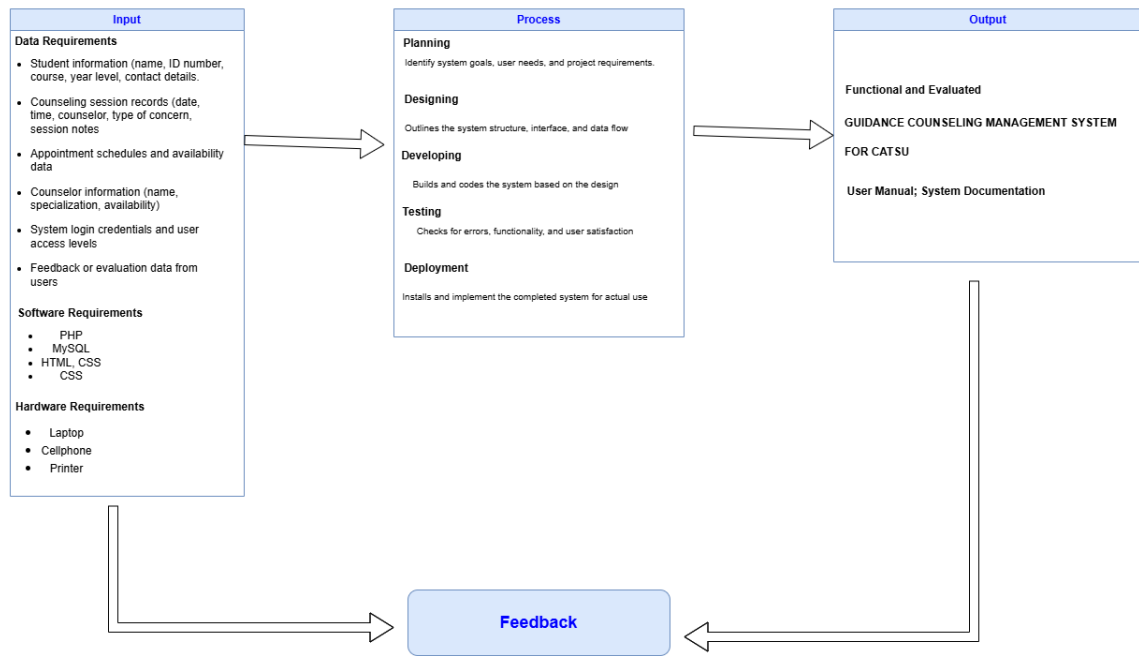


Figure 1. Conceptual Framework

This study is structured using the **Input-Process-Output (IPO) model**, which shows the logical flow of activities and resources throughout the development of the system. The **Input** consists of the necessary requirements such as student and counselor information, appointment schedules, and other related data, along with the software and hardware tools needed for system creation and operation. The **Process** includes the key stages of system development—**planning, designing, developing, testing, and deployment**—where each phase plays an important role in transforming inputs into a functional output. Planning identifies user needs and objectives; designing structures the system's interface and database; developing involves the actual coding and system

construction; testing ensures accuracy and usability; and deployment implements the system for actual use.

The **Output** of the framework is the fully developed **Guidance Counseling Management System**, which automates counseling appointment scheduling, digital record-keeping, and report generation. The system aims to reduce paperwork, minimize scheduling conflicts, and enhance the accessibility of counseling services to students. By establishing a systematic flow of processes through the IPO model, the framework ensures that every aspect of system development aligns with the main goal of the study — to improve the efficiency, organization, and responsiveness of guidance and counseling operations at CatSU.

CHAPTER 2

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter presents a review of related literature, studies, and technologies relevant to the development of the **Guidance Counseling Management System (GCMS)**. It discusses various concepts, models, and previous research that provide the foundation for understanding how guidance and counseling services can be improved through digital innovation. The review covers studies on the management and organization of guidance services, the integration of information and communication technologies in counseling systems, and the evaluation of web-based platforms that enhance accessibility and efficiency. By examining both local and international research, this chapter identifies the trends, similarities, and gaps in existing systems, which serve as the basis for developing a localized GCMS tailored to the needs of Catanduanes State University.

Guidance Counseling Management System

Kartadinata (2022) conducted a study titled *Quality Improvement and Management System Development of School Guidance and Counseling Services*, published in the **Journal of Universitas Negeri Malang**. The study developed and evaluated a model to improve the quality of guidance and counseling services across various school levels. By introducing a computerized management framework, the research highlighted the

significance of systematizing guidance services through structured procedures and digital recordkeeping. This aligns with the present study's objective to develop a centralized digital system that supports efficient management of counseling sessions and student information.

Fataroni (2023), in *Implementation of Guidance and Counseling Service Management in an Effort to Improve the Quality of Student Learning* from **Bale Literasi**, examined how the management functions of planning, organizing, leading, and controlling contribute to improving student learning outcomes. The study emphasized that proper management of guidance services enhances not only student performance but also counselor efficiency. This is closely related to the current study, which also focuses on improving management efficiency by automating routine tasks in the counseling office.

Similarly, Karunia et al. (2023) in *Management Strategies for Guidance and Counseling Services in Schools to Support Student Achievement*, published in **Manajia**, presented a literature-based analysis of effective management strategies for school guidance programs. Their work underscores the importance of well-defined processes, counselor coordination, and record organization. These insights support the rationale for developing a structured information system like the GCMS to ensure organized service delivery.

Furthermore, Jannah, Neviyarni, and Sukur (2022) in *The Role of Counselors in Developing Guidance and Counseling Management in Schools* from **Quality Journal**, explored how counselors contribute to designing and maintaining guidance management

systems. The authors found that counselor involvement in system development improves implementation and user acceptance. This supports the inclusion of counselors and staff feedback in the design process of the proposed GCMS, ensuring that the system meets actual user needs.

Technologies Integrated in GCMS

Nasution et al. (2024), in *Integration of Information Technology in Guidance and Counseling Management in Schools: Challenges and Implementation Strategies* published in **Attanwir Journal**, discussed the integration of IT into guidance counseling management. The study identified challenges such as limited digital literacy, technical constraints, and resource availability, but emphasized that proper training and infrastructure lead to more efficient service delivery. This finding supports the technological framework of the proposed GCMS, which integrates IT tools to simplify administrative tasks.

Majjate et al. (2024) proposed an *AI-Powered Academic Guidance and Counseling System Based on Student Profile and Interests*, published in **MDPI**. Their system utilized artificial intelligence to recommend personalized academic guidance. Although more advanced than the proposed GCMS, it demonstrates the growing trend of using intelligent

technologies to improve student support services. This reinforces the idea that automation and data-driven systems enhance the responsiveness and accuracy of guidance operations.

Azmin (2023), in *Integrating ICT into Career Counseling: An Empirical Study of the TACC Digital Content Model in Malaysian Secondary Education* from **HRMars**, investigated the integration of ICT in delivering career counseling services. The study found that digital models improve student engagement and counselor productivity. This supports the use of web technologies in your GCMS, which also aims to increase accessibility and user interaction.

In addition, Setyawan et al. (2023) in *Design of Continuous Web App: Guidance and Counseling Management Information System at SMKN 1 Purwakarta using Laravel Framework*, published in **Lembaga KITA**, developed a dynamic GCMS using the Laravel framework. The study's design approach and use of modern web tools closely parallel the methodology and technologies of your own system.

Likewise, Balmores et al. (2024) created an *Online Individual Inventory and Counseling Information System* for **Cagayan State University–Lasam**, which digitized student records and counseling forms. Their work demonstrated that a localized web-based system effectively streamlines information management and reduces manual documentation — an outcome that directly parallels the goals of your proposed GCMS.

Evaluation of Web-Based GCMS

Sudrajat et al. (2024), in *Evaluating Technology Acceptance in Counseling Services: An Analysis of Principal Indicators Utilizing the UTAUT2 Model via SEM-PLS*, published in EJournal UPI, examined factors affecting technology acceptance among counselors and administrators. The study concluded that performance expectancy and ease of use strongly influence user adoption of digital counseling platforms. This evaluation model can guide future assessments of your GCMS's usability and acceptance.

Guntara et al. (2023), in *Web-Based Counseling Skills Evaluation Information System Using Design Science Research Methodology (DSRM) Approach*, published in IJADIS, developed and assessed a system to evaluate counselor performance. Their use of the DSRM framework demonstrates how system development and evaluation can occur in iterative stages — similar to your project's methodological structure.

Lastly, Bato and Marcial (2016), in *Students' Attitudes Towards the Development of an Online Guidance Counseling System* from IITTL Journal, analyzed student perceptions toward web-based counseling platforms. Findings revealed positive attitudes toward digital accessibility and convenience. This supports the user-centered design of your GCMS, which aims to enhance student engagement through an online interface.

Synthesis

The reviewed studies collectively emphasize the growing importance of technology in improving the management and delivery of guidance and counseling services in educational institutions. Under the first thematic area, researchers such as Kartadinata (2022), Fataroni (2023), and Karunia et al. (2023) highlighted that effective management systems enhance counselor performance, improve service organization, and positively influence student learning. Their findings suggest that structured guidance management, when supported by clear planning and documentation, leads to more efficient and responsive services. Likewise, Jannah et al. (2022) underscored the crucial role of counselors in developing and maintaining such systems, pointing out that user involvement ensures better implementation and sustainability. Together, these studies show that the modernization of counseling services depends not only on management strategies but also on the active participation of those who use the system.

Studies under the second thematic area demonstrate that integrating technology into counseling operations brings substantial benefits but also introduces new challenges. Nasution et al. (2024) and Azmin (2023) both noted that information and communication technologies improve accessibility, accuracy, and efficiency in service delivery, while also requiring adequate training and infrastructure. Majjate et al. (2024) showed the potential of artificial intelligence to personalize counseling, while Setyawan et al. (2023) and Balmores et al. (2024) developed web-based systems using frameworks such as Laravel and PHP/MySQL to digitalize appointment scheduling and record management. These

studies illustrate the steady shift toward digital transformation in guidance services, proving that technological integration enhances both counselor workflow and student engagement.

Meanwhile, the third thematic area centers on the evaluation and user acceptance of web-based guidance systems. Sudrajat et al. (2024) applied the UTAUT2 model to examine user acceptance, showing that perceived usefulness and ease of use significantly influence adoption. Guntara et al. (2023) employed a design science research approach to evaluate system effectiveness and counselor performance, while Bato and Marcial (2016) explored positive student attitudes toward online counseling platforms. These studies confirm that web-based systems are generally well-received and effective when designed around user needs and tested through structured evaluation methods.

Despite these advancements, a noticeable **research gap** remains. Most existing systems focus on either specific counseling functions or the evaluation of technology use but rarely combine both development and implementation in a localized academic context. Few studies address the unique challenges of integrating an all-in-one digital platform for managing counseling schedules, student cases, and reports within a single institution. The present study addresses this gap by developing a comprehensive, web-based **Guidance Counseling Management System (GCMS)** tailored for Catanduanes State University. Unlike previous works, it unifies administrative management, appointment scheduling, and documentation processes into one accessible system—providing an innovative, student-centered solution that modernizes the university’s guidance operations.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the technical background of the study, which includes the hardware, software, data, and user requirements needed for the development and operation of the **Guidance Counseling Management System (GCMS)** for Catanduanes State University. These components serve as the technical foundation of the system, ensuring that it functions effectively and meets the needs of its intended users.

Hardware Requirements

The hardware components provide the physical infrastructure necessary to develop, test, and deploy the GCMS. Adequate hardware ensures smooth operation, reliable performance, and accessibility for both developers and users in the Guidance and Counseling Office.

Table 1. Hardware Requirements

Hardware Requirement	Specification	Description
Laptop	Intel Core i5-1335U (13th Gen), 8GB RAM, 512GB SSD, 15.6" FHD, Wi-Fi 6E, Windows 11, Cypress Green	Used by developers and staff for system development, testing, and operation. Ensures smooth performance and multitasking..

Printer	Epson L3110, Print/Scan/Copy functions	Used to print reports, forms, and other official documents from the system.
Router	PLDT Home Fiber, Wireless LAN	Provides network connectivity for accessing the GCMS via the internet or intranet.

The hardware requirements listed above provide the necessary resources for system development and implementation. The laptop or desktop computer serves as the main platform for programming, debugging, and testing. The printer allows counselors to generate and print reports and other documentation, while the router ensures stable internet connectivity, enabling users to access the web-based GCMS anytime and anywhere within the school network.

Software Requirements

The software requirements consist of the programs, tools, and environments used for coding, designing, testing, and deploying the system. These software components ensure that the GCMS operates efficiently and can be accessed via any compatible web browser.

Table 2. Software Requirements

Software Requirement	Specification	Description
Operating System	Windows 11	Manages computer operations and coordinates processing of system applications.
Web Server	Apache	Hosts and runs the GCMS locally for development and testing.
Database Management System	MySQL	Stores and manages data including student profiles, counselor information, and appointment records.
Programming Language	PHP, HTML, CSS, javascript	Used for developing system functionalities, web design, and interactive features.
Web Browser	Google Chrome	Used to access and test the GCMS interface and functionality.
Office Tools	Microsoft Office	Used for documentation and preparation of reports.
Localhost Environment	Xampp	Provides an integrated platform for local web server deployment and testing.

The software requirements are essential for developing a stable and interactive web-based platform. XAMPP provides a local server environment where the system can be developed and tested using Apache and MySQL. The programming languages PHP, HTML, CSS, and JavaScript ensure that the GCMS is functional, visually appealing, and user-friendly. Web browsers such as Chrome and Firefox allow the team to test the

responsiveness and compatibility of the system, while office tools are used for documentation and reporting.

Data Requirements

The data requirements identify the specific information that the GCMS will store, process, and manage. These data sets are crucial for providing accurate records, generating reports, and supporting decision-making within the Guidance and Counseling Office.

Table 3. Data Requirements

Construct/ Variables	Source of Data	Data Gathering Technique	Instrument	Data Processing
Counselor Officials Information	Guidance Office	Document Analysis	Documentary Review	Input and validation in the database
Student Profile	Student	Interviews and Surveys	Questionnaire	Structured data input in the system
Appointment Schedule	Guidance Office	Observation / Interview	Interview Guides	Data encoding and automated tracking
Counseling Records	Guidance Office	Interviews	Structured Form	Digital documentation and reporting
System Evaluation	Users (students, counselors, admin)	Descriptive Survey Method	Questionnaire	Mean and frequency count for analysis

The table outlines the essential data collected and processed by the system. Counselor and student information form the core of the GCMS database, allowing efficient tracking of sessions and appointments. Appointment schedules and counseling records ensure that each student interaction is recorded systematically. The evaluation data gathered through surveys and interviews provide valuable feedback that helps improve the system's usability and functionality.

User Requirements

The user requirements define the system functionalities and access privileges for each type of user administrators, counselors, and students. These requirements serve as the blueprint for the GCMS design and ensure that each user role performs its intended tasks efficiently.

Table 4. User Requirements

Module	Function	Description	User Role
Dashboard	Overview of System Status	Displays key metrics such as number of active users, appointment reports	Admin
Appointment Management	Schedule and Track sessions	Allows admin to set manage and track appointments	Admin
Student Management	View and Update Student Records	Add update or remove student records	Admin
Counseling Management	Manage student counseling Information	Organizing student counseling information	Admin, Counselor

SMS Notification	Send and Receive Notification	Send and receive for counselors from client	Admin, Counselor
Generate Report	Counseling	Produce reports based on session	Admin, Counselor

The table shows the user requirements needed for the system, a digital platform intended to streamline how students access and receive support. For students, its easy ways to book appointments and manage their counselling journey, for counsellor and staff, it provides a central, secure place to organized student profile, keep track of session notes, and generate important reports, replacing the hassle of manual records-keeping. The system also includes a dashboard for administrator, offering a quick and clear overview of counselling activities and the overall effectiveness of the service.

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter discusses the design, development, and methodology applied in creating the Guidance Counseling Management System (GCMS) for Catanduanes State University. It details the requirements analysis, documentation, design of software and systems, development, testing, respondents, and statistical tools used for system evaluation. The software development methodology applied in this project is the **Agile Methodology**, which emphasizes iterative development, collaboration with stakeholders, and continuous improvement. Agile was selected because it allows flexibility in addressing user feedback and ensures that the system is aligned with the needs of the Guidance Counseling and Testing Office (GCTO).

Requirements Analysis

The requirements were gathered through multiple techniques, including interviews with guidance counselors, surveys with students, and direct observations of existing counseling processes at CatSU. The current process is largely paper-based, requiring students to manually fill out consent forms and book counseling appointments through walk-ins. This creates inefficiencies such as scheduling conflicts, missed appointments, and difficulties in record management.

The proposed system addresses these issues by introducing a digital platform for booking appointments, managing counseling records, and generating reports. The data collected through requirement analysis provided the foundation for both functional and non-functional requirements

Requirements Documentation

The requirements for the Guidance Counseling Management System are presented in tabular form below, categorized into functional, non-functional, assumptions, and constraints, each followed by a discussion.

Table 1. Functional Requirements

REQUIREMENTS	DESCRIPTIONS
Student Login	Students must be able to, log in, and securely access the system.
Appointment Scheduling	Students must be able to book, cancel, or reschedule counseling sessions.

Counselor Case Management	Counselors must manage appointments, view student profiles, and update case notes.
Administrative Management	Administrators must manage user accounts, oversee system usage, and generate reports.
Notifications and Reminders	The system must send automated confirmations, reminders, and updates to students and staff.

These functional requirements ensure that the core features directly support the daily activities of students, counselors, and administrators. The ability to register, schedule, and receive notifications streamlines communication and reduces scheduling errors, while administrative management enhances oversight and reporting.

Table 2. Non-Functional Requirements

REQUIREMENTS	DESCRIPTIONS
Usability	The system should have a user-friendly and intuitive interface for both students and staff.

Security	All data must be encrypted, with role-based access to ensure privacy and confidentiality.
Reliability	The system must function consistently with minimal downtime during office hours.
Performance	The system should load pages and process requests within three seconds.

Non-functional requirements define the quality attributes of the system. Usability ensures ease of navigation, while security safeguards sensitive data. Reliability minimizes disruptions in service, and performance standards ensure a smooth and efficient user experience.

Design of Software, Systems, Product and/or Processes

Context Diagram

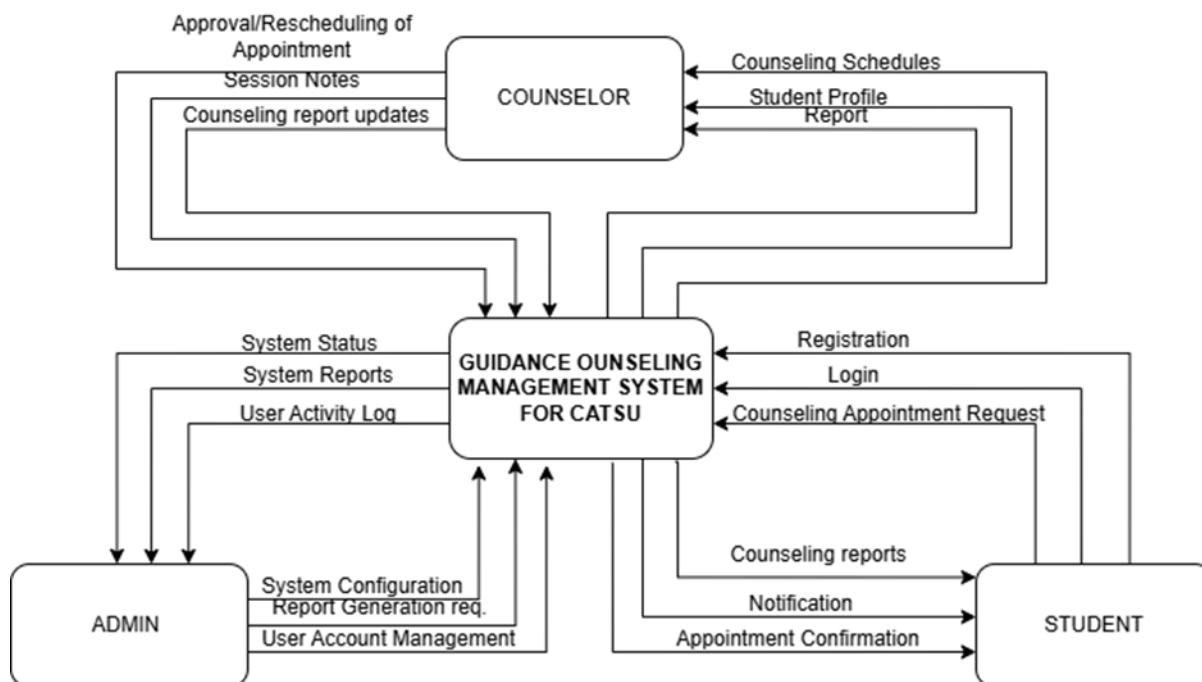


Figure2 Context Diagram

The context diagram of the Guidance Counseling Management System (GCMS) for Catanduanes State University illustrates the system as a single process and shows its interaction with three primary external entities: the Student, the Counselor, and the Admin. This diagram establishes the system boundary and clarifies how data flows between the users and the system, thereby providing a high-level view of its functionality.

The Student is one of the key actors in the system. Students interact with the GCMS by registering or logging in, managing their personal information, and submitting

counseling requests. They may also select or reschedule appointments based on their availability and receive important system outputs such as appointment confirmations, reminders, and counseling reports. This interaction ensures that students can conveniently access guidance services without relying on manual forms or walk-in requests, making the process more efficient and student-centered.

The Counselor represents the professional users of the system. Counselors access the GCMS to view submitted appointment requests, approve or adjust schedules, and conduct counseling sessions. After each session, they input session notes, progress updates, and recommendations into the system, which are stored as counseling records. They also have the ability to generate reports summarizing student progress and session outcomes. By digitizing these tasks, the system reduces the paperwork load of counselors and allows them to focus more on delivering effective guidance to students.

The Admin serves as the system overseer. Administrators log in to the GCMS to manage user accounts, configure system settings such as scheduling policies or notification rules, and monitor system activity through logs and generated reports. They are responsible for ensuring that the system operates smoothly and securely, while also using the aggregated data for decision-making and policy development. The system provides admins with outputs such as user activity logs, statistical reports, and summaries of counseling trends, which are essential for evaluating the performance of the guidance services.

The arrows in the diagram represent the data flows between each external entity and the system. Inputs include login credentials, appointment requests, counselor notes,

and configuration data, while outputs consist of confirmations, notifications, reports, and monitoring feedback. By clearly showing these exchanges, the context diagram highlights how the GCMS acts as a centralized platform that integrates the needs of students, counselors, and administrators.

In summary, the context diagram emphasizes that the GCMS is designed to streamline the overall counseling process at CatSU by reducing manual workloads, ensuring accurate record-keeping, and providing timely access to guidance services. It portrays the system as the central hub where all users interact, ultimately improving the efficiency and effectiveness of guidance counseling operations within the university.

DATA FLOW DIAGRAM

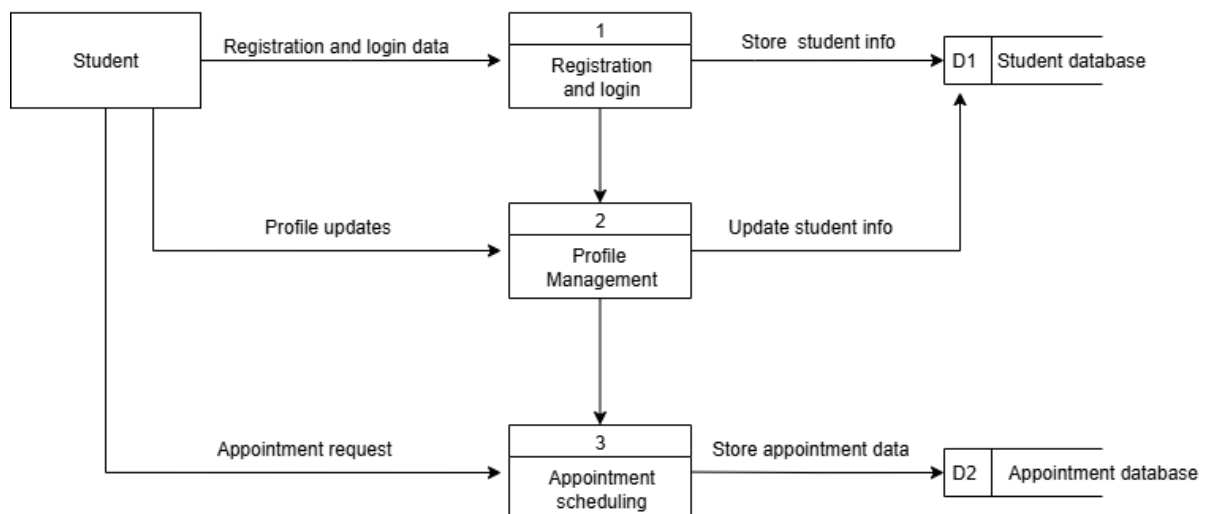


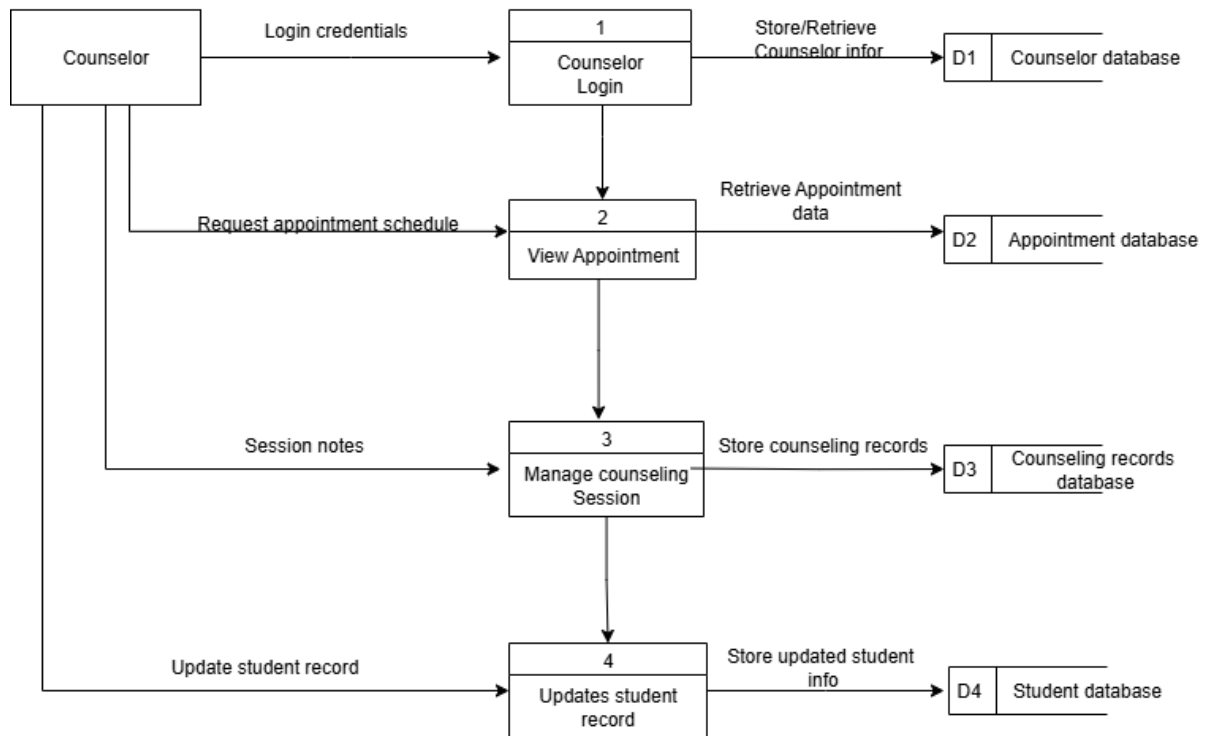
Figure3. Data Flow Diagram Level 1 (Student)

The Data Flow Diagram (DFD) Level 1 for the student illustrates how learners interact with the Guidance Counseling Management System (GCMS) in carrying out their essential tasks. As an external entity, the student communicates with the system through three main processes: registration and login, profile management, and appointment scheduling. Each of these processes ensures that student-related data is properly handled and securely stored in the appropriate databases.

When students first access the system, they begin with the registration and login process. In this stage, the information they provide such as personal details during registration or login credentials during authentication is processed by the system and stored in the Student Database. This establishes the student's identity and enables secure access to the system's features.

After successfully logging in, students may update or edit their personal information through the profile management process. Any changes they make, such as updating contact details or academic-related information, are processed by the system and saved again in the Student Database. This guarantees that the records are always accurate and up to date.

Students also make use of the appointment scheduling process, which allows them to request, reschedule, or cancel counseling sessions. The details they provide, including the date, time, and purpose of the appointment, are stored in the Appointment Database. By managing this information digitally, the system ensures that appointments are efficiently tracked and can be easily accessed by both students and counselors.



Data Flow Diagram Level 1 (Counselor)

The Data Flow Diagram (DFD) Level 1 for the counselor presents the key activities that a guidance counselor performs when using the Guidance Counseling Management System (GCMS). As an external entity, the counselor interacts with the system primarily to log in, view appointments, manage counseling sessions, and update student records. These processes ensure that counselors are able to perform their responsibilities effectively, while the system keeps the information organized and secure.

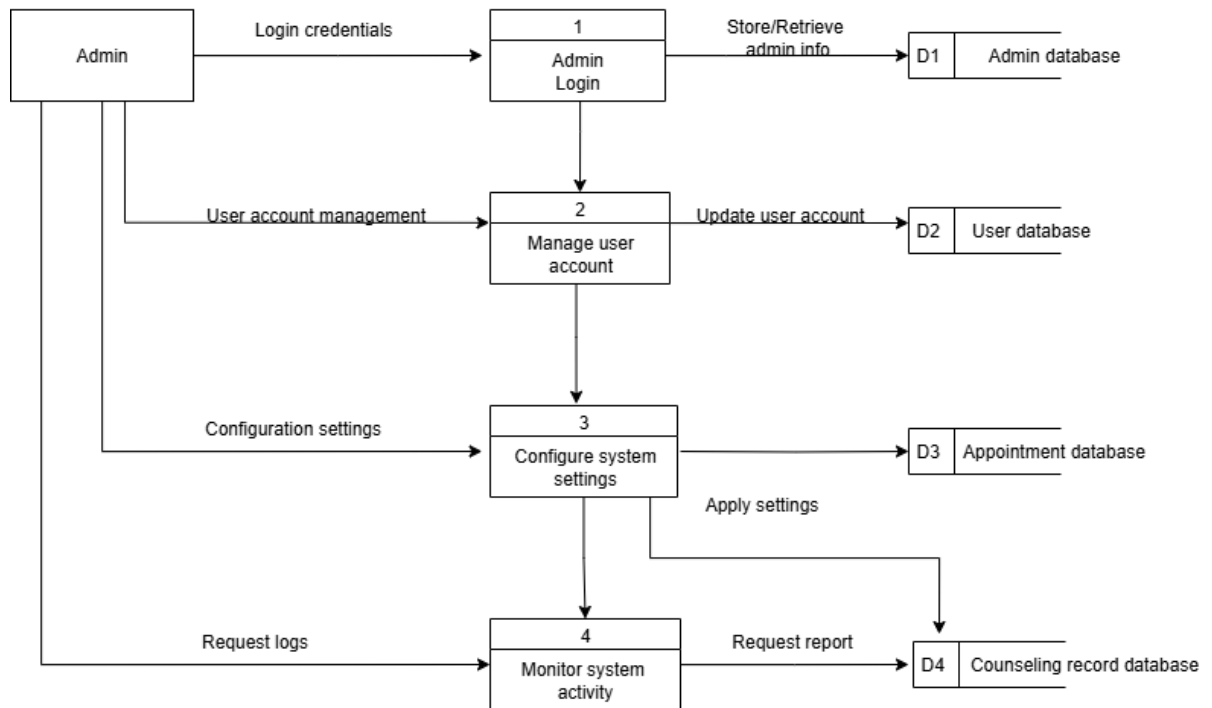
The interaction begins with the counselor login process, where the counselor provides authentication credentials to access the system. The information is validated and

stored in the Counselor Database, ensuring that only authorized personnel can use the system's features.

Once logged in, counselors can access the view appointments process, which allows them to check their scheduled sessions with students. The system retrieves and processes appointment information stored in the Appointment Database, providing counselors with a structured view of upcoming and pending appointments. This feature supports proper time management and reduces scheduling conflicts.

Another important process is managing counseling sessions, where counselors document the outcomes of each session, including session notes, recommendations, and observations. This data is processed and securely stored in the Counseling Records Database, enabling continuity of service and proper tracking of a student's guidance history.

Finally, counselors are able to perform student record updates, which involve modifying or verifying student details when necessary. The updates made through this process are stored in the Student Database, ensuring that student profiles remain accurate and consistent.



Data Flow Diagram Level 1 (Admin)

The Data Flow Diagram (DFD) Level 1 for the administrator outlines the core responsibilities of the admin user in the Guidance Counseling Management System (GCMS). The administrator serves as the system overseer, ensuring that the platform operates smoothly, user accounts are properly managed, and system settings are appropriately configured. In this diagram, the administrator interacts with the system through three primary processes: login, user account management, and system configuration and monitoring.

The process begins with admin login, where the administrator submits authentication credentials to access the system. These details are validated against the

Admin Database, which ensures that only authorized system administrators are able to access and control the platform. This process provides security and prevents unauthorized changes to the system's operations.

After successful login, the administrator can carry out user account management, which includes creating new accounts, updating user information, or deactivating accounts when necessary. The information processed during this activity is stored in the User Database, ensuring that students, counselors, and other system users have properly managed accounts. This process maintains user integrity and role-based access within the GCMS.

The third process involves system configuration and monitoring, where administrators adjust system-wide settings such as notification preferences, appointment policies, and other operational rules. These configurations are stored in both the Appointment Database and the Counseling Records Database to align with scheduling and record-keeping functions. By doing so, the system ensures that the workflows remain consistent with institutional policies and operational requirements.

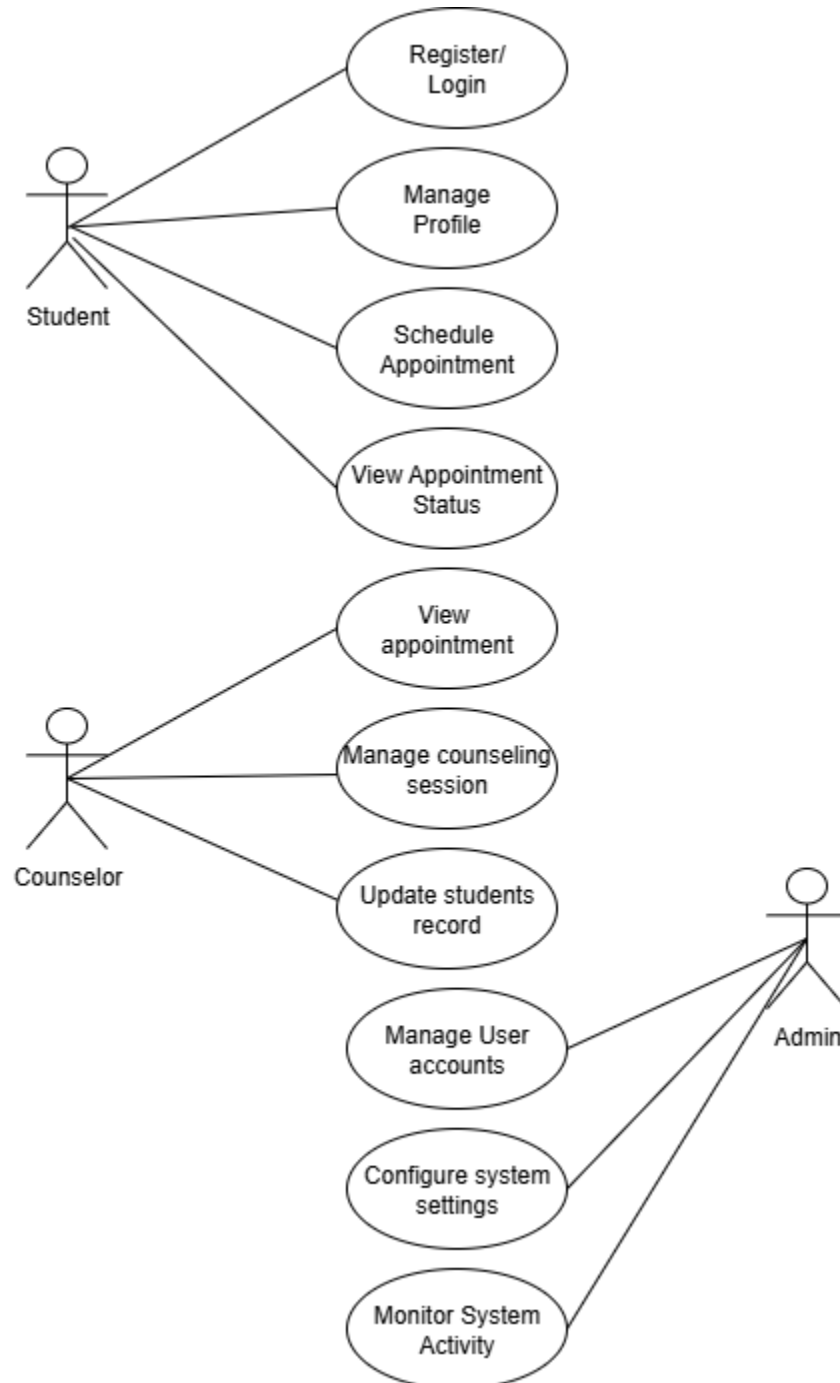
USE CASE DIAGRAM

Figure4 Use Case Diagram

The Use Case Diagram for the Guidance Counseling Management System (GCMS) illustrates how the three main actors **Student, Counselor, and Admin** interact with the system. Each actor performs specific tasks that reflect their role in the counseling process, and all interactions are contained within the boundaries of the system.

The **Student** plays a primary role in the GCMS as the main recipient of guidance services. Students begin by registering and logging into the system, which grants them access to the platform's features. Once inside, they can manage their personal profiles to ensure that their information is accurate and up to date. One of the core functions for students is **scheduling appointments** with their assigned counselor. If needed, students also have the ability to **reschedule or cancel appointments**, ensuring flexibility in managing their sessions. Additionally, students can view the status of their appointments to confirm schedules and monitor any updates made by the counselor.

It starts with the system making things easy by allowing the person to, fill out consent forms, and schedule their intake interview. Once that's done, the GCTO Staff steps in to review and approve their request, ensuring everything is in place. If approved, the Guidance Counselor takes over, sitting down with the person to listen and understand their needs during an intake interview. Together, they begin the counseling sessions. At each step, decisions are made collaboratively, keeping the person's well-being in mind. The process adapts based on their progress, either continuing counseling or concluding if their needs are met. It's a thoughtful, supportive journey with everyone working together.

The Counselor also logs into the system to securely access their account and perform counseling-related activities. After authentication, counselors are able to view scheduled appointments, which helps them organize their sessions and manage their time effectively. During or after each session, counselors use the system to record counseling notes, feedback, and recommendations, ensuring that the student's progress is properly documented. Counselors are also responsible for updating student records, which keeps profiles current and accurate for future reference.

The Admin acts as the overseer of the system. Similar to other users, the admin begins with the login process, which ensures secure access to system management tools. Once logged in, the admin can perform tasks such as managing user accounts, which involves creating, updating, or deactivating accounts for students and counselors. In addition, the admin is responsible for configuring system settings, such as defining scheduling rules or notification preferences, to ensure that the platform operates according to institutional policies. Another important role of the admin is to monitor system activity, which allows them to oversee user operations and ensure the integrity and security of the platform.

Activity Diagram/Process Swimlane

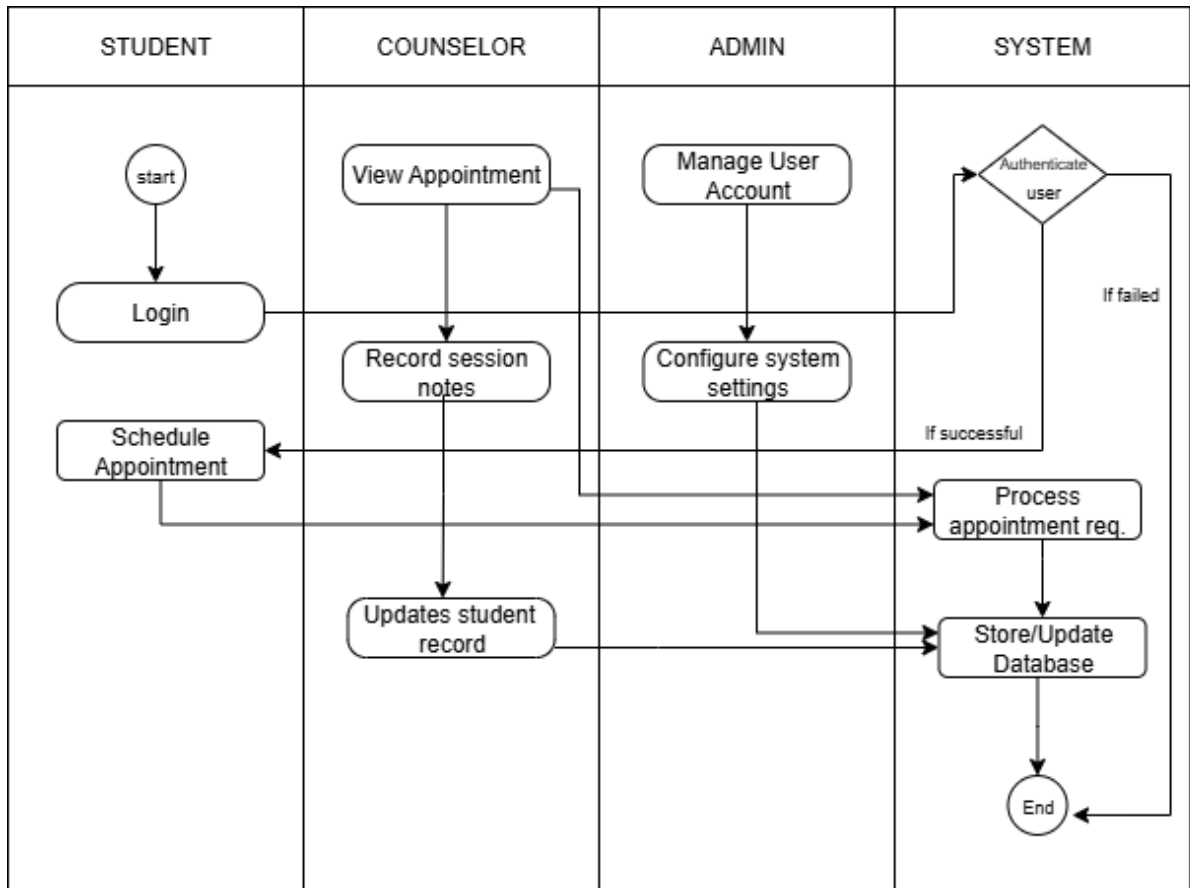


Figure5 Activity Diagram/Process Swimlane

The figure shows the swimlane diagram of the Guidance Counseling Management System, which illustrates the responsibilities and flow of interactions among the three main actors: Student, Counselor, and Admin, together with the System. Each lane defines the role of a particular actor, while the arrows demonstrate the sequence of processes until all actions are finalized and stored by the system in the database.

For the Student, the process begins with login or registration. The System authenticates the user's credentials, and if authentication fails, an error message is displayed, thereby terminating the process. If authentication is successful, the student proceeds to schedule or reschedule an appointment. The System processes this request and updates the counselor's appointment list. The counselor can then view the scheduled appointments through the system, after which the System stores and updates the appointment information in the database.

For the Counselor, the process also begins with login. The System authenticates the counselor's credentials, and upon successful login, the counselor can access and view the appointments. The counselor may then record session notes and update student records after counseling sessions. These updates are received by the System, which stores the information into the appropriate databases, such as counseling records and student information, ensuring that data is accurate and securely maintained.

For the Admin, the process starts with login as well. The System verifies the admin's credentials, and upon successful authentication, the admin can perform administrative functions, including managing user accounts and configuring system settings. All changes made by the admin are automatically recorded by the System and stored in the user accounts and configuration databases.

Across all roles, whether Student, Counselor, or Admin, every flow converges into one unified ending: the System storing or updating data in the central database. This highlights the role of the Guidance Counseling Management System as the central handler

of information, ensuring that all actions are securely processed and maintained. The process concludes once all interactions are completed and finalized by the system.

Entity-Relationship Diagram

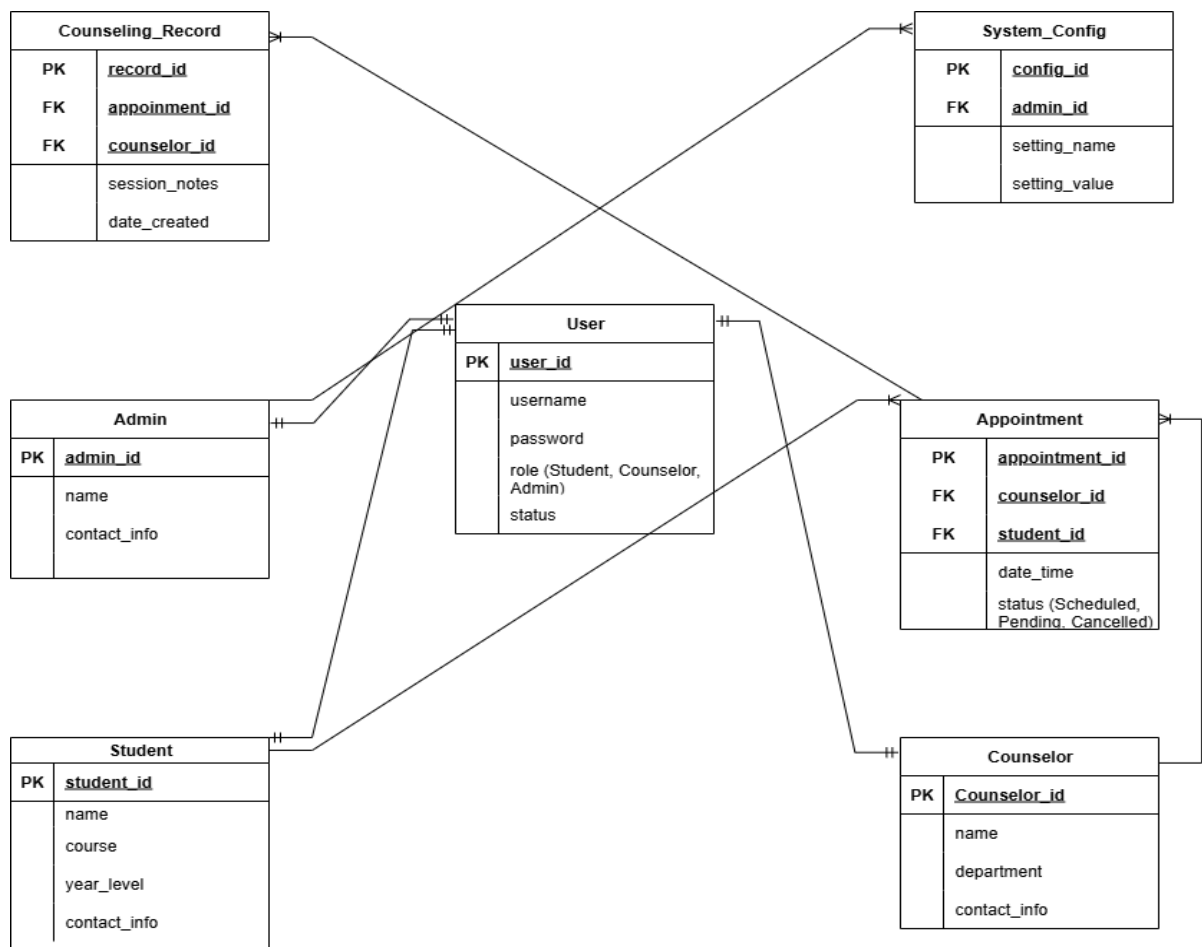


Figure6 Entity-Relationship Diagram

The figure shows the **Entity Relationship Diagram (ERD)** of the **Guidance Counseling Management System**, which illustrates how data entities are related and managed within the system. The diagram highlights the logical structure of the database, defining the entities, their attributes, and their interrelationships.

At the center of the diagram is the **Appointment** entity, which serves as the core of the system. It connects the **Student** and the **Counselor**, establishing the relationship between those who request guidance services and those who provide them. A student can create multiple appointments, while a counselor can handle multiple appointments. Each appointment contains details such as the date, time, and status, which allow both students and counselors to track counseling schedules.

The Counseling Record entity is directly linked to the appointment. It stores the session notes and outcomes recorded by the counselor after each counseling session. This ensures that the system maintains a history of student counseling sessions, which can be referenced for follow-up and future decision-making.

The Student entity contains personal and academic details such as student number, course, year level, and contact information. Each student is linked to their user account and is able to request one or more appointments with counselors. Similarly, the Counselor entity holds counselor-specific details such as name, department, and contact information. Counselors manage multiple appointments and are responsible for updating counseling records.

The Admin entity is also represented in the diagram, with the primary responsibility of managing user accounts and system configurations. This is supported by the System Configuration entity, which stores configurable settings maintained by the administrator. While the admin does not directly interact with the appointments, they play a crucial role in ensuring that the system operates smoothly and securely.

Overall, the ERD emphasizes that the Appointment entity is the central hub of the system, linking students and counselors, while other entities such as Counseling Records, User Accounts, and System Configurations support the complete functionality of the Guidance Counseling Management System.

SYSTEM ARCHITECTURE

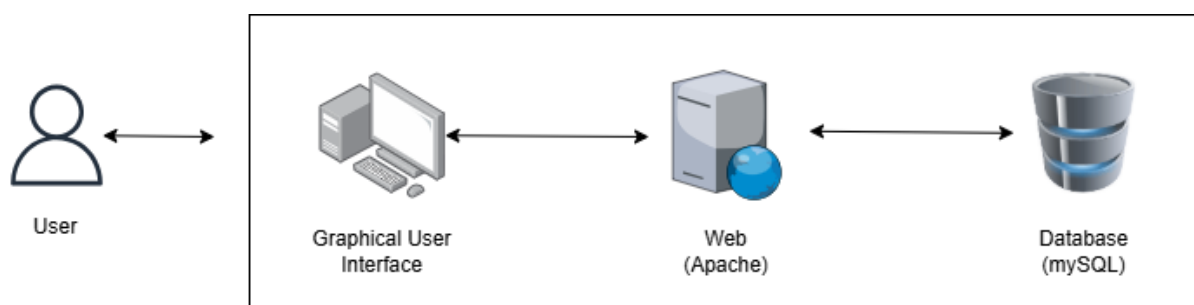


Figure 7 System Architecture

The system architecture of the **Guidance Counseling Management System (GCMS)** follows a three-tier structure that ensures a seamless, secure, and efficient interaction between users and the application's core functionalities. At the forefront of this

architecture are the users, who may take on different roles such as **students, counselors, or administrators**. These users interact with the system through a **Graphical User Interface (GUI)**, which serves as the visual and interactive layer of the application. Through this interface, users can perform a range of tasks such as registering an account, updating their profile, booking or rescheduling counseling appointments, and accessing assigned counselor sessions.

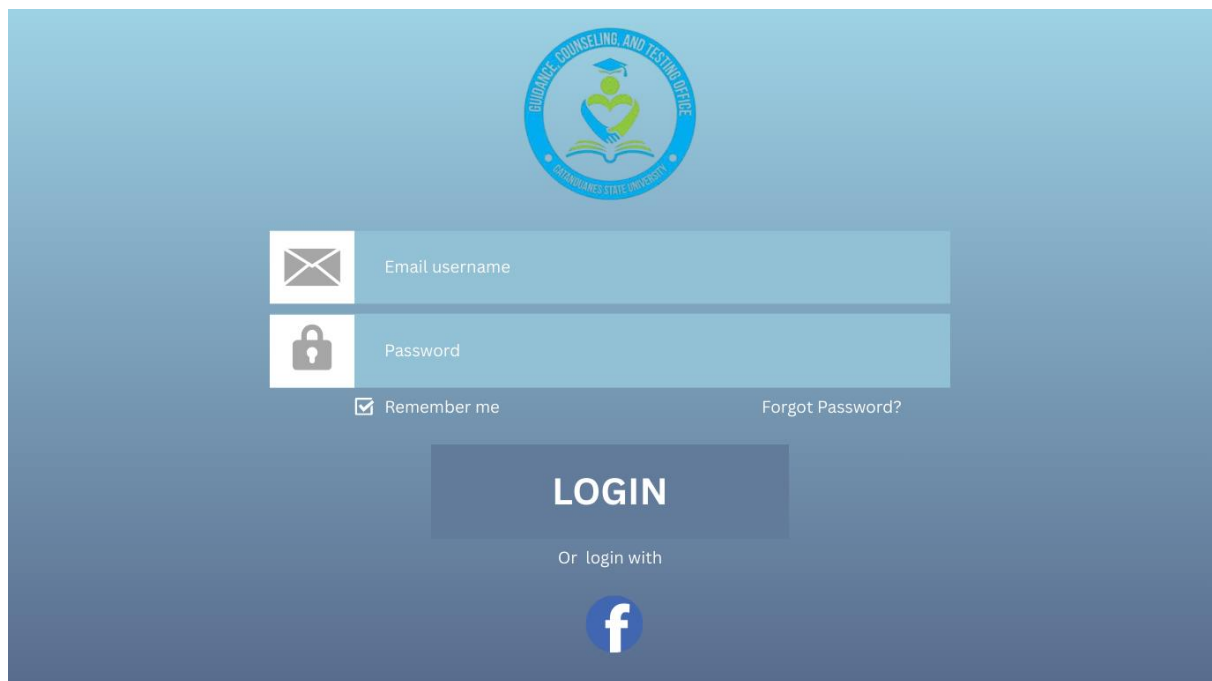
When a user initiates an action, the GUI captures the input and communicates it to the **web server**, which in this case is powered by **Apache**. The web server functions as the intermediary between the user interface and the database. It processes incoming requests, executes the appropriate logic, and determines what kind of response or data is required. For example, if a student wants to view their scheduled counseling session, the web server interprets this request, applies the necessary business rules, and prepares a query for the database.

The final tier of the architecture is the **database**, built on **MySQL**, which acts as the central repository for all information in the system. This includes user account details, counseling schedules, student records, counselor session notes, notifications, and administrative data. The web server queries the database to either retrieve or update this information based on the user's interaction. Once the operation is completed—whether it is fetching scheduled appointments or saving a new counseling record—the database sends the result back to the server, which then communicates the outcome to the GUI for the user to see.

This three-tier architecture ensures that the GCMS operates with efficiency, scalability, and security, while maintaining a clear separation of concerns between the **presentation layer (GUI), application logic (web server), and data management (database).**

SYSTEM PROTOTYPE

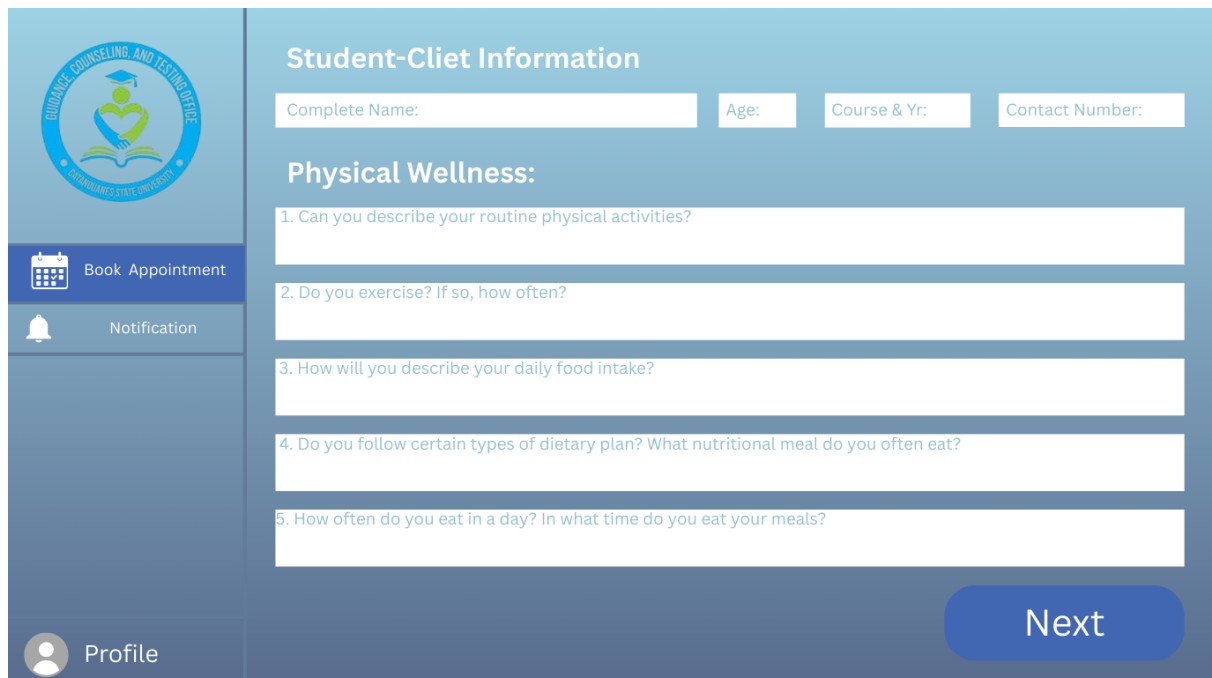
Student Side



The image shows a login page prototype for the Student Side of the GCMS. The page has a blue gradient background. At the top center is a circular logo for 'GUIDANCE, COUNSELING, AND TESTING OFFICE' at 'CHANGUNGWANG STATE UNIVERSITY'. Below the logo are two input fields: 'Email username' with an envelope icon and 'Password' with a lock icon. Below the password field is a 'Remember me' checkbox and a 'Forgot Password?' link. A large 'LOGIN' button is centered below these fields. Below the button is the text 'Or login with' followed by a Facebook 'f' logo.

Figure8 Login Page

The login page is designed as the system's secure access point, where users are required to enter their email username and password to verify their identity before accessing the system. It provides additional features such as a 'Remember Me' option for easier future logins, a 'Forgot Password' link for account recovery, and an alternative login method through Facebook integration, ensuring both security and convenience for users.



The screenshot displays a web interface for a student-client information form. On the left is a vertical sidebar with a blue header containing a circular logo for 'GUIDANCE, COUNSELING, AND TESTING SERVICE' at 'SRIJAYAWARDHANA STATE UNIVERSITY'. Below the logo are three menu items: 'Book Appointment' (with a calendar icon), 'Notification' (with a bell icon), and 'Profile' (with a person icon). The main content area has a light blue header with the title 'Student-Client Information'. Below this, there are four input fields for 'Complete Name:', 'Age:', 'Course & Yr:', and 'Contact Number:'. A section titled 'Physical Wellness:' follows, containing five numbered questions, each with a corresponding text input field:

1. Can you describe your routine physical activities?
2. Do you exercise? If so, how often?
3. How will you describe your daily food intake?
4. Do you follow certain types of dietary plan? What nutritional meal do you often eat?
5. How often do you eat in a day? In what time do you eat your meals?

 At the bottom right of the main area is a large blue button labeled 'Next'.



GUIDANCE, COUNSELING, AND TESTING REFERENCE
NNAMDI AZIKIWE STATE UNIVERSITY

 Book Appointment

 Notification

 Profile

Physical Wellness:

6. How many hours do you sleep in a day?

7. What is the time of your waking up and sleeping hour?

8. How would you describe your sleep (sound and good sleep or shallow sleep)?

9. Do you smoke, drink, or take drugs? If so, how often and since when?

Intellectual Wellness:

1. What are your strengths in terms of your knowledge, abilities, and skills?

2. How do you use these strengths in your daily tasks?

Next



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NNAMDI AZIKIWE STATE UNIVERSITY

 Book Appointment

 Notification

 Profile

Intellectual Wellness:

3. Describe your academic performance?

4. How do you enhance your knowledge, abilities, and skills?

5. If there are potentials that still need to be enhanced, what are those?

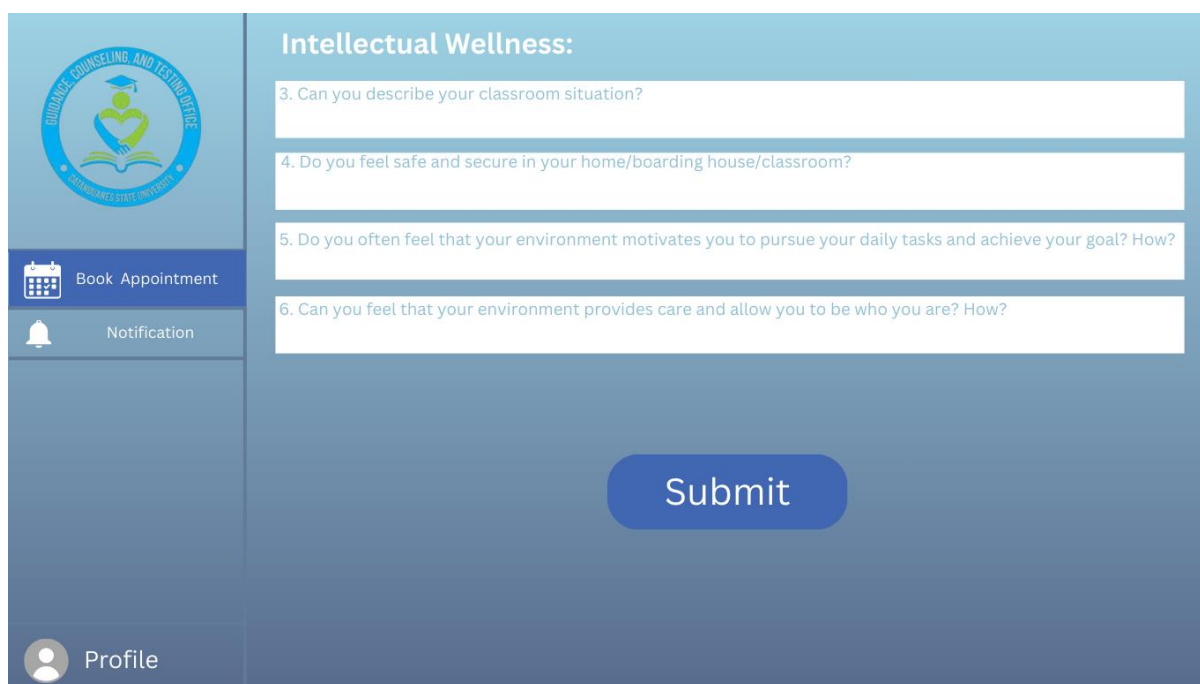
6. Is there weakness that you would like to be improved? What weakness?

Environmental Wellness:

1. How will you describe your home recently?

2. How will you describe your boarding house recently?

Next



Intellectual Wellness:

3. Can you describe your classroom situation?

4. Do you feel safe and secure in your home/boarding house/classroom?

5. Do you often feel that your environment motivates you to pursue your daily tasks and achieve your goal? How?

6. Can you feel that your environment provides care and allow you to be who you are? How?

Submit

Profile

Figure9 Booking Page

This page serves as the Student-Client Information form, where students are required to provide their personal details such as complete name, age, course and year, and contact number. It also includes a section on Physical Wellness, Intellectual Wellness and Environmental Wellness that gathers important health-related information through guided questions about physical activities, exercise habits, daily food intake, dietary plans, and meal frequency. The form helps the Guidance, Counseling, and Testing Office collect relevant data for assessing the student's overall wellness and providing appropriate support or counseling services.

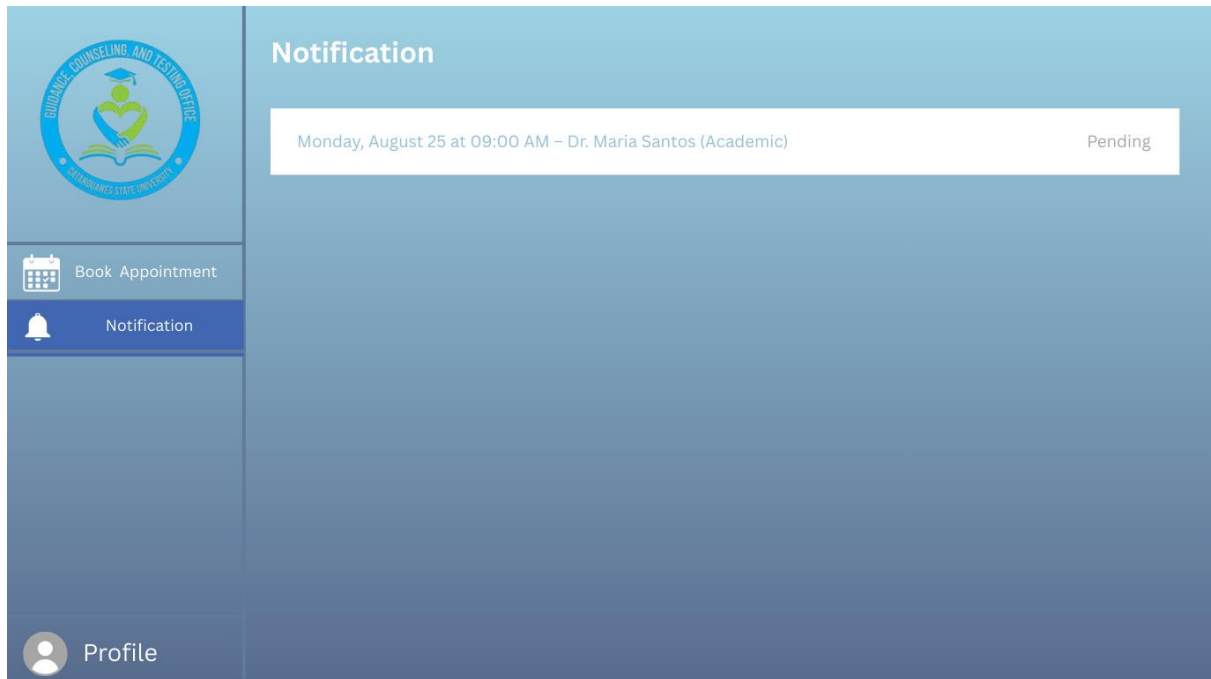
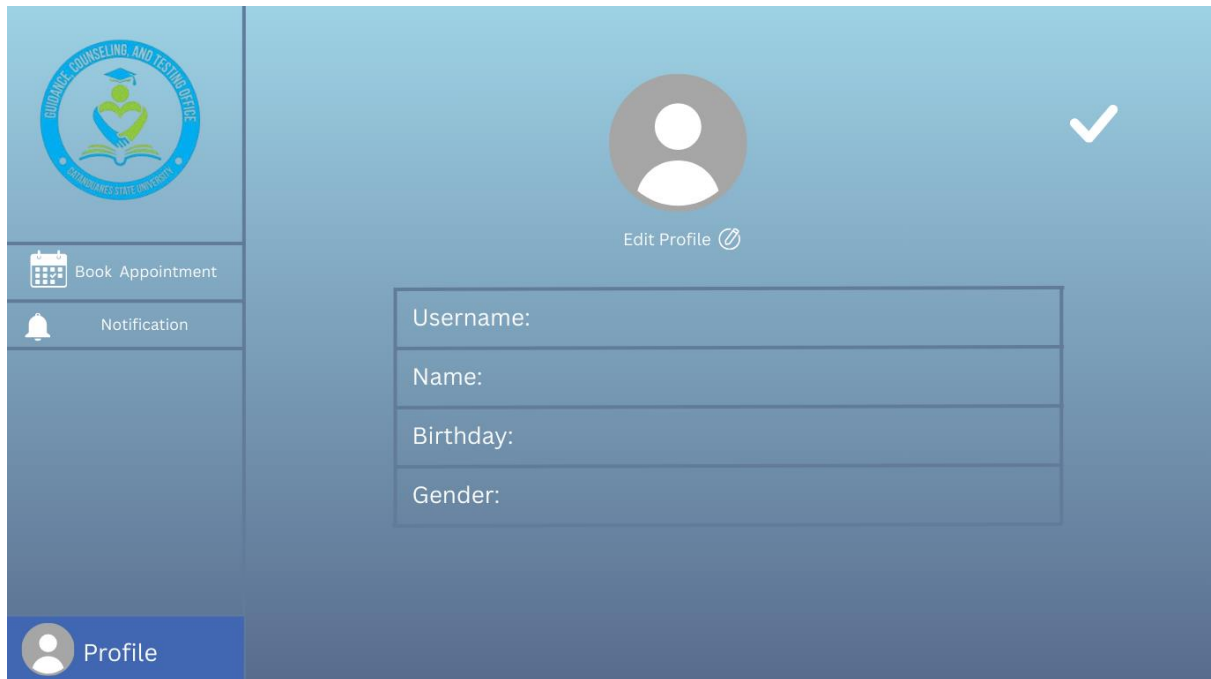


Figure10 Notification Page

The Notification Page displays important updates, reminders, and announcements from the Guidance, Counseling, and Testing Office. It serves as a communication hub where students can be promptly informed about scheduled appointments, upcoming events, counseling reminders, or any other relevant notices to ensure they stay updated and engaged with the services provided by the system.



The screenshot displays a user profile page with a blue gradient background. On the left is a vertical sidebar containing the university's logo at the top, followed by buttons for 'Book Appointment' (with a calendar icon) and 'Notification' (with a bell icon). At the bottom of the sidebar is a 'Profile' button with a person icon. The main content area features a large circular profile picture placeholder, an 'Edit Profile' link with a pencil icon, and a confirmation checkmark. Below these are four stacked text input fields labeled 'Username:', 'Name:', 'Birthday:', and 'Gender:'.

Figure11 Profile Page

The Profile Page allows students to view and manage their personal information within the system, including their name, birthday, gender. It serves as a centralized space where users can update their information when necessary, ensuring accuracy for counseling records and appointment management.

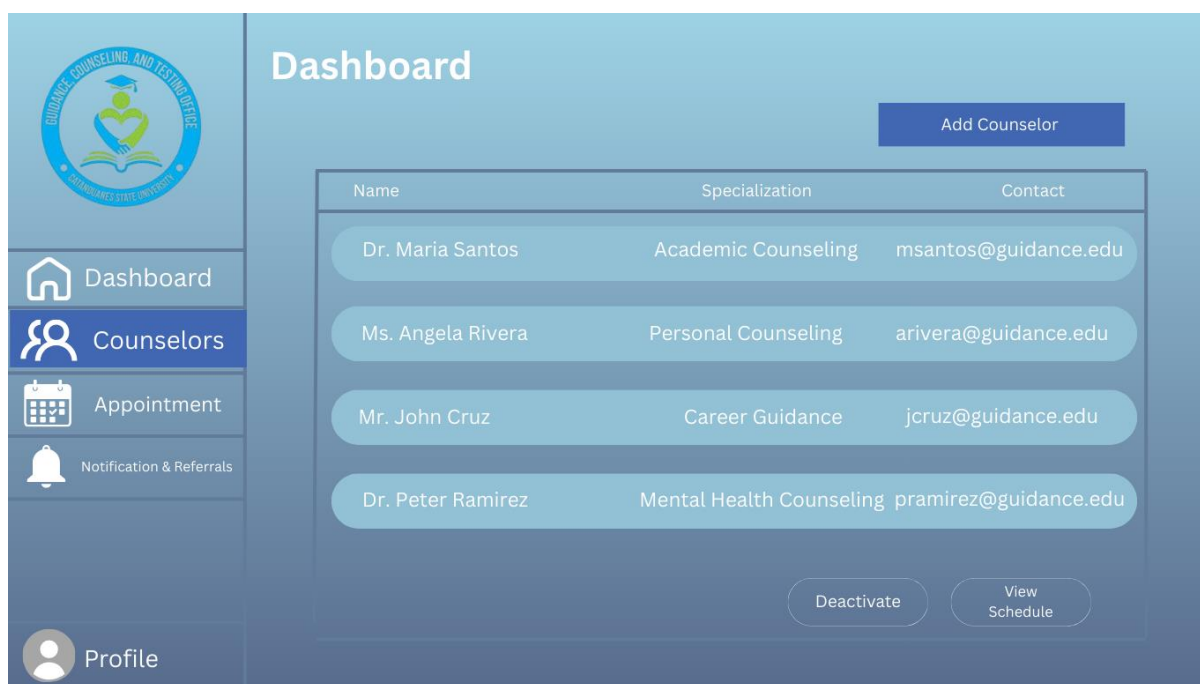
Admin Side



Figure13 Admin dashboard

The **Guidance Counseling Management System** dashboard serves as the central interface of the platform, providing administrators with a comprehensive overview of counseling operations. It highlights essential statistics such as the total number of appointments, the current number of active counselors, and pending referrals. In addition, it enables efficient management of counselor information, including profiles, contact

details, and status updates, ensuring streamlined monitoring and improved delivery of guidance and counseling services.



Guidance, Counseling, and Testing Office
SAN DIEGO STATE UNIVERSITY

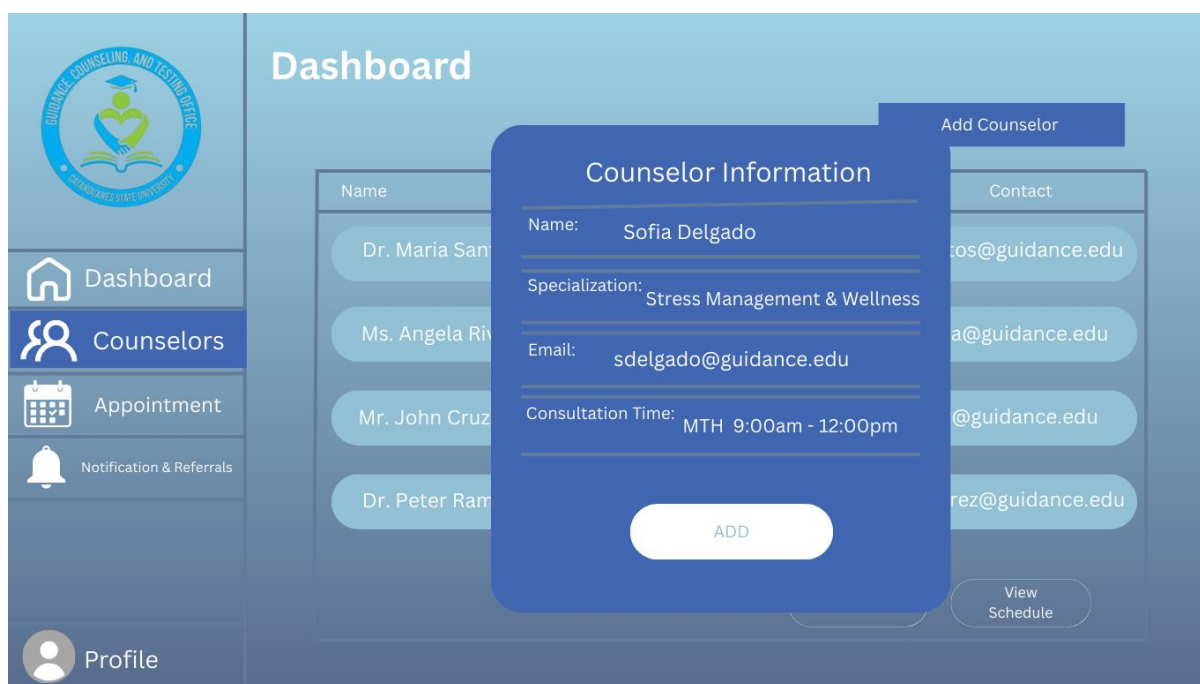
Dashboard

[Add Counselor](#)

Name	Specialization	Contact
Dr. Maria Santos	Academic Counseling	msantos@guidance.edu
Ms. Angela Rivera	Personal Counseling	arivera@guidance.edu
Mr. John Cruz	Career Guidance	jcruz@guidance.edu
Dr. Peter Ramirez	Mental Health Counseling	pramirez@guidance.edu

[Deactivate](#) [View Schedule](#)

[Dashboard](#) [Counselors](#) [Appointment](#) [Notification & Referrals](#) [Profile](#)



. Figure14 Counselors Page

The **Counselors Page** of the Guidance Counseling Management System serves as a dedicated space for managing counselor information, where administrators can view a complete list of counselors along with their names, specializations, and contact details. It provides essential functions such as adding new counselors, updating profiles, deactivating accounts, and accessing individual schedules, making it easier to organize assignments and ensure that students are referred to the right counselor based on their specific needs.



. Figure15 Appointment Page

The Manage Appointment Page of the Guidance Counseling Management System provides administrators with an organized view of all scheduled counseling sessions, listing the date, time, counselor, specialization, and assigned student for each appointment. This page streamlines the process of handling requests by allowing administrators to approve, decline, or reschedule sessions, ensuring proper coordination between students and counselors and promoting a more efficient and well-structured counseling service.

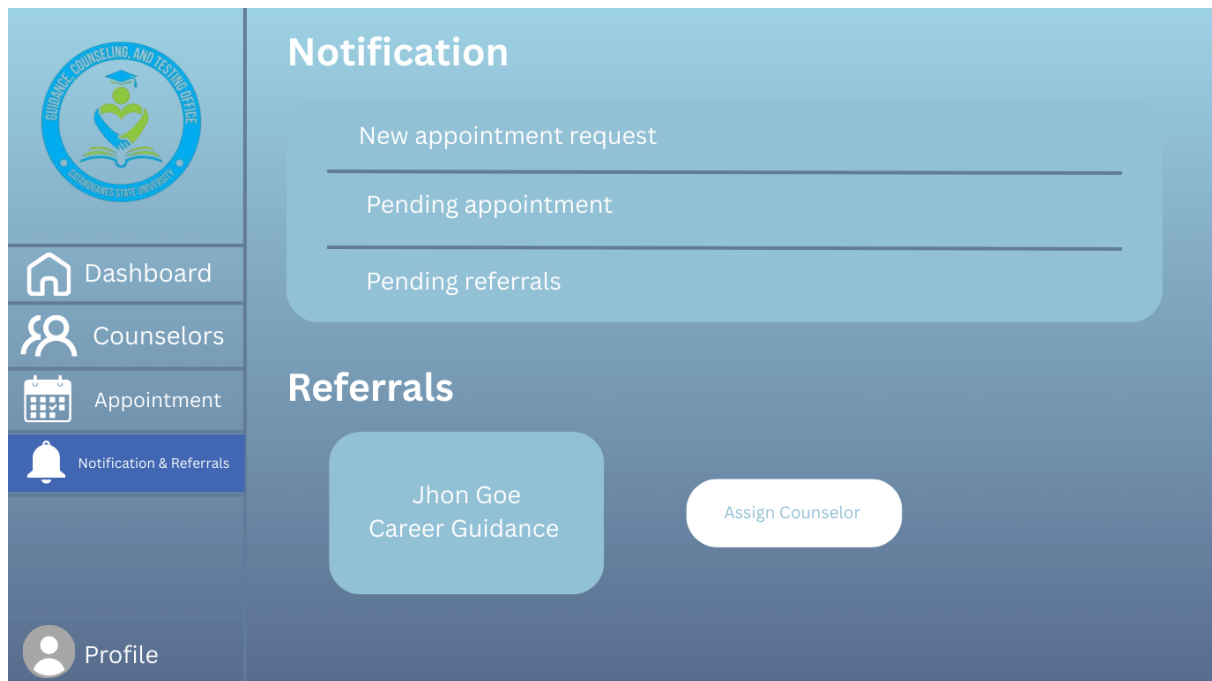


Figure16 Notification and Referrals Page

The **Notification & Referrals Page** of the Guidance Counseling Management System serves as the communication hub where administrators and counselors can track updates, reminders, and pending tasks related to counseling activities. It provides timely alerts about new appointments, schedule changes, and important announcements while also managing student referrals for specialized counseling services. Through this feature, administrators can easily monitor the flow of referrals, ensure that students are directed to the appropriate counselor based on their needs, and maintain smooth coordination within the guidance office.

Counselors Side

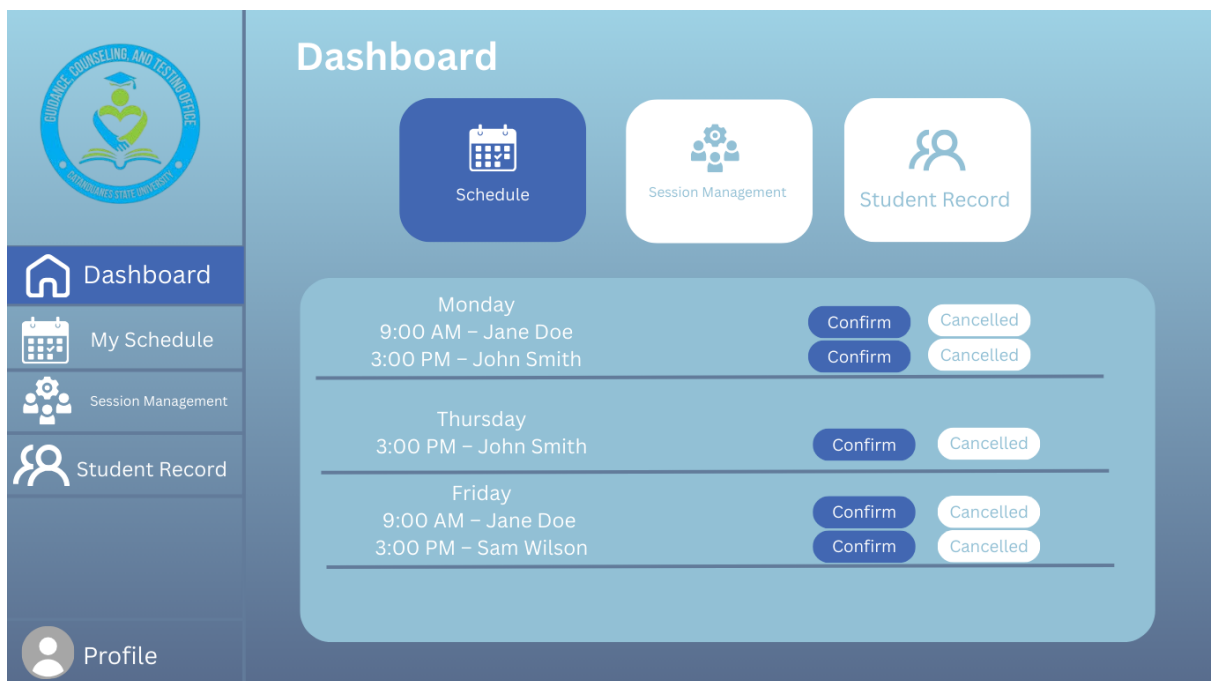


Figure17 Counselor dashboard

The Dashboard serves as the central control panel of the counseling system, providing users with quick access to the different features such as schedule management, session management, and student records. It displays a clear and organized list of upcoming counseling appointments arranged by day and time, including the names of the students assigned, and provides the counselor with the option to either confirm or cancel

each session, making it easier to manage daily activities and monitor the overall flow of appointments.

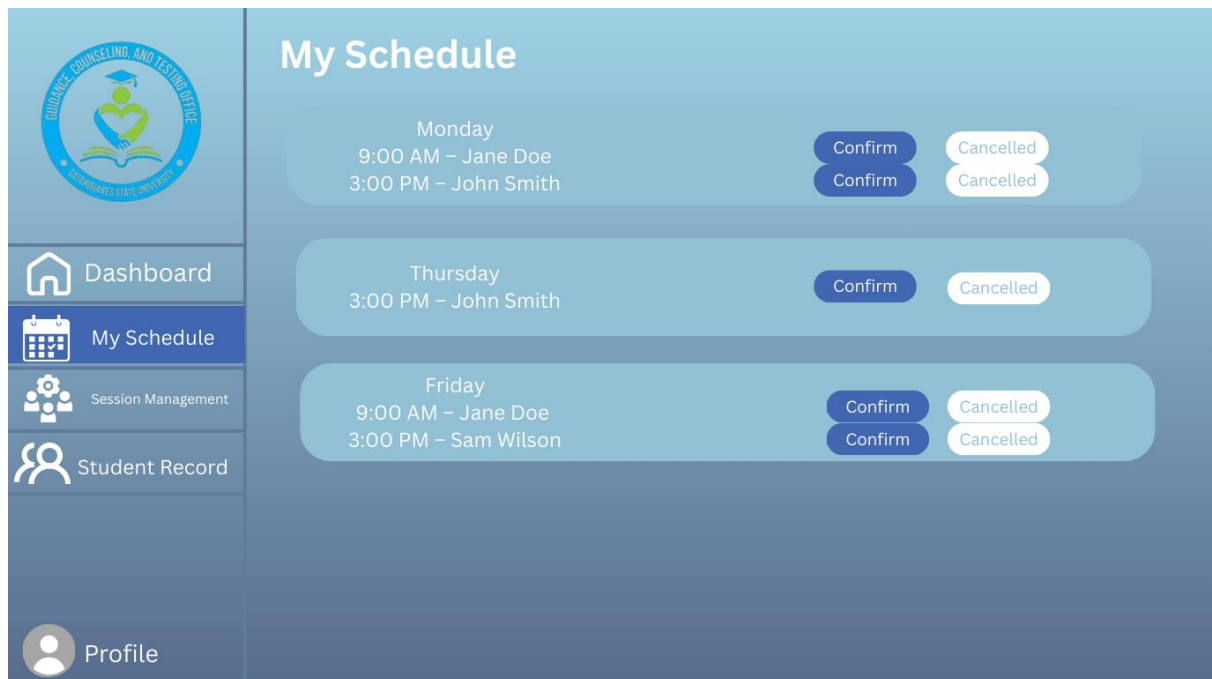


Figure18 Schedule Page

The My Schedule section serves as a personalized appointment manager for counselors, allowing them to view their assigned counseling sessions organized by day and time. It presents the list of students scheduled for counseling, along with options to confirm attendance or cancel sessions when necessary. This feature helps counselors efficiently plan their availability, avoid scheduling conflicts, and maintain a well-structured timetable for delivering counseling services.



GUIDANCE, COUNSELING, AND TESTING OFFICE
SAN JUAN COLLEGE

Dashboard

My Schedule

Session Management

Student Record

Profile

Session Management

Active Appointment

Student Name: Maria Lopez
Year & Course: 2nd Year, BS Nursing
Session Type: Academic Counseling
Status: Ongoing

Previous Appointment

Student Name: Daniel Cruz
Year & Course: 3rd Year, BS Accountancy
Session Type: Stress Management /
Personal Counseling
Status: Completed

Student Name: Angela Torres
Year & Course: 4th Year, BS Education
Session Type: Career Guidance
Status: Completed

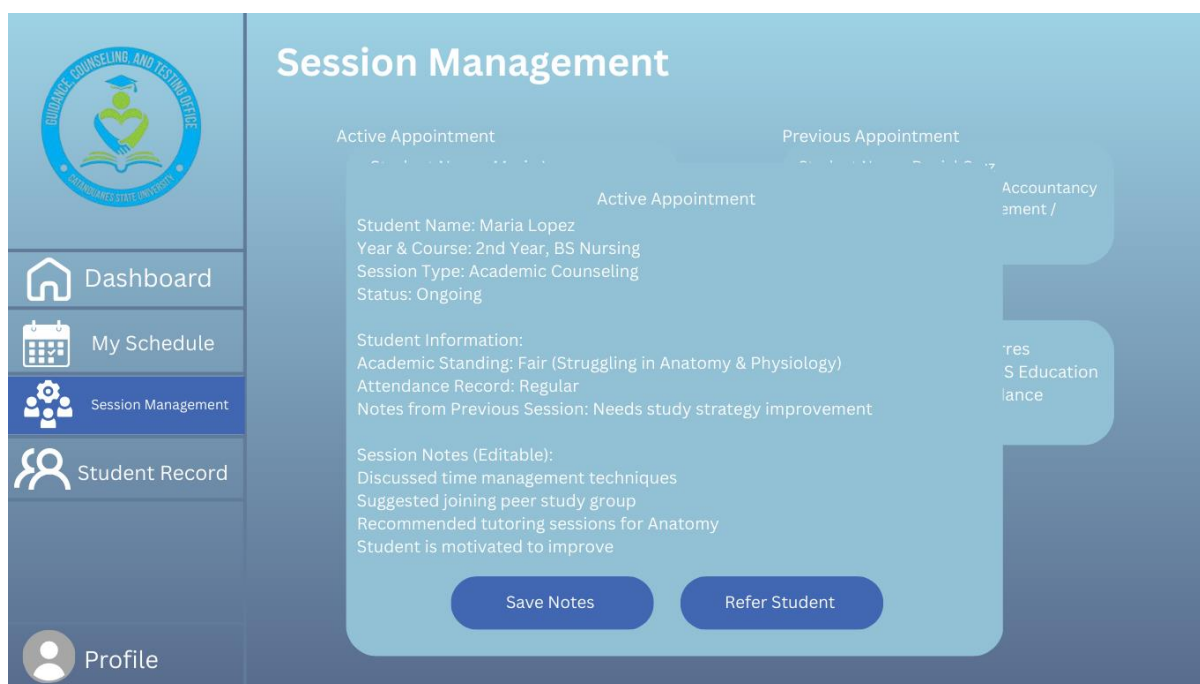


Figure19 Sessions Page

The Session Management system serves as a comprehensive platform for counselors to oversee and document student appointments by displaying both active and previous sessions with details such as student name, year and course, type of counseling, and status, while also offering an in-depth view of active appointments that includes the student's academic standing, attendance record, and notes from prior sessions, along with editable session notes where counselors can record discussions on study strategies, time management, and recommended interventions, and features action options like saving notes or referring the student for further guidance.

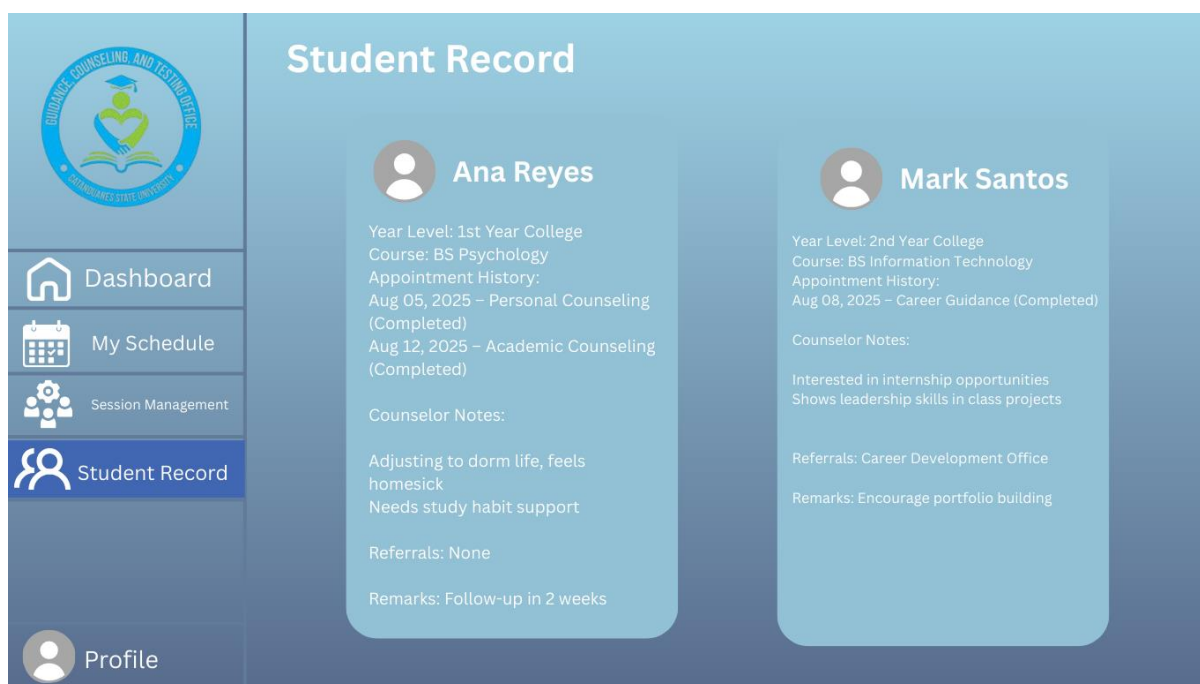


Figure20 Student Record Page

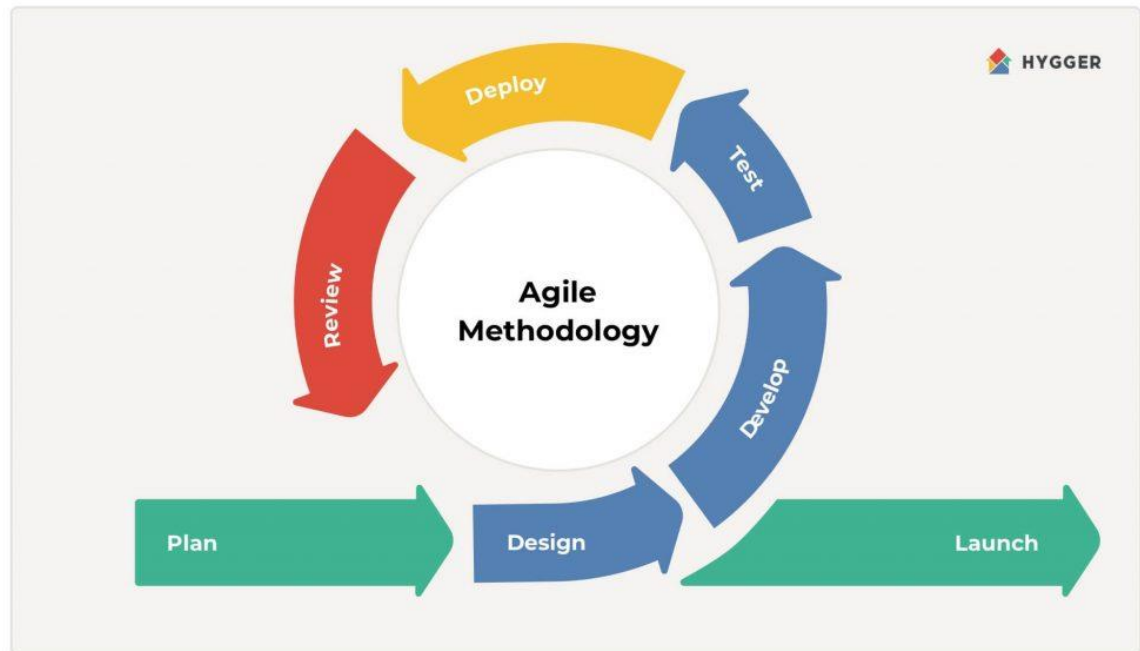
The Student Record section contains comprehensive information about individual students, which includes their year level, course, and complete appointment history indicating the type and date of counseling sessions they have attended. It also provides detailed counselor notes that reflect the student's needs, concerns, and progress, along with any referrals made to other departments or offices and remarks for future follow-up. This feature ensures that counselors have an updated and holistic view of each student's development, making it easier to give appropriate guidance and consistent support.

Development and Testing

Software Development Method

For the development of the **Guidance Counseling Management System (GCMS)**, the researchers adopted the **Agile Methodology** to ensure that the system was built in an iterative and user-centered manner. Unlike traditional linear models, Agile allowed flexibility in refining the system features based on the feedback of its intended users—students, counselors, and administrators of CatSU. Through incremental sprints, the essential modules such as appointment scheduling, counselor session management, and administrative account control were developed, tested, and improved until they met the identified requirements of the study.

In addition to development, this chapter also focuses on the **evaluation of the GCMS**. To assess the system's quality and acceptability, the researchers employed the **Descriptive Survey Method** guided by the **ISO/IEC 25010 software quality model**. This framework measures software through eight key characteristics, including functionality, usability, reliability, and security. The evaluation results provided a clear indication of how well the system addressed the needs of its users and whether it could be adopted effectively within the university.



ImageSource: <https://hygger.io/guides/wp-content/uploads/2021/04/what-is-agile-1024x614.jpg>

Figure21 Agile Methodology

The development of the **Guidance Counseling Management System (GCMS)** was carried out using the **Agile Methodology**. Unlike traditional linear models, Agile emphasizes flexibility, collaboration, and incremental progress. This methodology was chosen because it allowed the researchers to continuously refine the system based on user feedback from students, counselors, and administrators. Agile ensured that the GCMS evolved according to the real needs of its end-users while reducing risks associated with rigid, one-time development.

At the start of the development process, the researchers conducted requirement gathering and user story creation. Through interviews, surveys, and consultations, the team identified the challenges in the existing manual guidance counseling process. These requirements were translated into user stories such as: *“As a student, I want to schedule or reschedule an appointment so that I can meet my counselor at a convenient time”* and *“As a counselor, I want to record session notes so that I can maintain accurate student records.”* These user stories served as the foundation for prioritizing features in each Agile iteration or sprint.

The planning phase involved selecting the most essential features for the first sprint. High-priority functions such as user authentication, appointment scheduling, and counselor viewing of appointments were chosen because they formed the core of the system’s operations. This allowed the researchers to deliver a working prototype early in the development process, which could immediately be tested and evaluated by potential users.

During the development phase, the system was built incrementally. The team worked on one sprint at a time, developing modules such as Student Registration and Login, Appointment Scheduling, Counselor Session Management, and Admin User Account Management. After each sprint, the partially completed system was demonstrated to users for feedback. This approach provided the flexibility to make changes based on suggestions, such as refining the user interface to be more intuitive or adjusting the flow of appointment notifications.

Testing was integrated into every sprint, ensuring that features were validated as soon as they were developed. Unit testing checked the correctness of individual modules, while integration testing ensured that modules worked together seamlessly. For example, the process of scheduling an appointment was tested end-to-end to confirm that the request made by a student was properly displayed on the counselor's dashboard and stored in the database. Bugs identified during testing were immediately corrected within the sprint before moving on to the next set of tasks.

The review and retrospective phase followed each sprint. In this phase, feedback was gathered from the stakeholders who interacted with the system. Students evaluated whether the scheduling feature was convenient, counselors checked the usability of session record management, and administrators tested the account management functions. The researchers discussed what went well, what challenges were encountered, and what improvements were needed in the next sprint.

Finally, the deployment phase was conducted after several completed sprints resulted in a functional and stable version of the GCMS. The system was deployed for trial use at CatSU, allowing real users to interact with it in an actual environment. Maintenance activities followed, focusing on minor refinements and enhancements based on additional feedback. Since Agile is iterative by nature, the system remains open for continuous improvement and expansion even after deployment.

Through the use of Agile Methodology, the development of the GCMS was made more adaptable and user-centered. The iterative sprints ensured that the system's features

were tested and validated continuously, reducing risks of failure. More importantly, Agile enabled active participation of students, counselors, and administrators throughout the development process, ensuring that the final product was aligned with their needs and expectations.

Evaluation

The evaluation of the Guidance Counseling Management System (GCMS) was conducted to measure its overall quality, usability, and acceptability among its intended users. To accomplish this, the researchers employed the Descriptive Survey Method, which made use of a structured questionnaire distributed to students, counselors, and administrators who were identified as the primary users of the system. The survey instrument was developed based on the ISO/IEC 25010 software quality model, which provides an internationally recognized framework for assessing software quality.

The evaluation focused on the eight characteristics outlined in ISO/IEC 25010. Functional suitability was assessed to determine whether the GCMS delivered the necessary features expected by its users, such as student appointment scheduling, counselor record management, and administrative control of accounts. Performance efficiency measured the responsiveness of the system in processing login requests, scheduling transactions, and updates to the database. Compatibility referred to the ability of the system

to operate seamlessly across different devices and web browsers commonly used in the university.

Usability was another crucial aspect, which examined the system's user interface, navigability, and ease of use for first-time users and frequent users alike. The reliability of the GCMS was assessed in terms of its ability to consistently perform its intended functions without system crashes or data errors, particularly during multiple simultaneous transactions. Security was evaluated to ensure that the system provided adequate protection for sensitive information, preventing unauthorized access to student and counselor records.

In addition, maintainability was measured to determine how easily the GCMS could be updated or modified when improvements or corrections were necessary. Finally, portability was considered in terms of the system's ability to adapt to different environments or be deployed in other academic institutions with minimal adjustments.

The results of the survey were tallied, analyzed, and interpreted to establish the system's overall acceptability. Based on user responses, the GCMS was found to be highly functional and reliable, with a user-friendly interface that met the needs of its intended users. Areas for improvement were also identified, particularly in enhancing system performance efficiency and strengthening security measures. These findings served as the basis for future refinements of the system.

Data Gathering Procedure

The study utilized both qualitative and quantitative approaches in gathering data to support the development and evaluation of the Guidance Counseling Management System (GCMS). During the development phase, data were collected through interviews, observations, and reviews of existing documents from the Guidance Office of CatSU. These activities provided the researchers with a clear understanding of the current manual process of managing appointments, counseling records, and student information. The insights gained from this stage allowed the researchers to identify the problems faced by students, counselors, and administrators, which were then translated into system requirements and user stories that guided the iterative development under the Agile methodology.

For the evaluation phase, the study employed a survey questionnaire anchored on the ISO/IEC 25010 software quality model. The instrument was validated by experts and administered to a sample group consisting of students, counselors, and administrators who tested the system during its trial deployment. The questionnaire aimed to measure the GCMS based on eight quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Respondents rated each item using a Likert scale, and the gathered data were tallied, analyzed, and interpreted to determine the overall quality and acceptability of the system.

Through this two-pronged data gathering approach, the study was able to capture both the requirements for development and the metrics for evaluation, ensuring that the

GCMS was not only responsive to user needs but also aligned with international software quality standards.

Respondents of the Study

The respondents of this study were selected to ensure that the data gathered would be accurate and relevant to the objectives of the research. They include three groups within the school community: students, counselors, and administrative staff. Each group was chosen because of their direct involvement and varying perspectives that could contribute meaningful insights to the study. By including these respondents, the research aimed to capture diverse viewpoints and experiences.

In total, the population of possible respondents consisted of 27 individuals. From this number, a sample of 9 participants was drawn to represent each group proportionally. Specifically, 5 students, 3 counselors, and 1 administrative staff member were included in the study. This distribution ensures that the gathered data reflects a balanced representation of the target population. The summary of respondents is presented in the table below.

Table 5. Respondents of the Study

Types of Respondent	Total Population	Sanple
Student	20	5

Counselors	5	3
Admin	2	1
	27	9

To effectively evaluate the objectives of the study, it was necessary to gather data from different groups within the school community. The respondents were composed of students, counselors, and administrative staff, with a total population of twenty-seven (27) individuals. From this population, a purposive sample of nine (9) participants was selected, consisting of five (5) students out of twenty (20), three (3) counselors out of five (5), and one (1) administrative staff member out of two (2).

This distribution was designed to ensure proportional representation of each group. By including individuals with distinct roles and responsibilities, the study was able to obtain diverse perspectives and insights that contributed to a more comprehensive understanding of the research problem. Such representation strengthens the reliability and validity of the findings, making them more reflective of the actual population being studied.

Statistical Tool

In analyzing the data collected from respondents, the study employed descriptive statistical tools to assess the system's usability, performance, and overall user

satisfaction. These tools were selected for their effectiveness in summarizing feedback in a clear and quantifiable manner, thereby supporting evidence-based evaluation of the developed system.

Table 6. Liker Scale

Range	Quantitative Equivalent	Qualitative Equivalent
4.21 – 5.00	5	Strong Agree
3.41 – 4.20	4	Agree
2.61 – 3.40	3	Neutral
1.81 – 2.60	2	Disagree
1.00 – 1.80	1	Strongly Disagree

Arithmetic Mean (Average)

$\sum X$ = Sum of all observed values

N = Number of observations or respondents

One of the primary tools used was the weighted mean ($\bar{X} = \sum X / N$). This statistical method was applied to compute the average score for each evaluation criterion based on the respondents' ratings. It allowed the researchers to determine the overall sentiment toward specific aspects of the system, such as functionality, interface design,

and communication features. The study also utilized frequency counts to determine how often particular responses or ratings appeared in the survey results. This helped identify trends and patterns in user feedback, particularly in areas where multiple respondents shared similar experiences or perspectives. To measure subjective responses, the study adopted a five-point Likert scale, which ranged from "Strongly Disagree" to "Strongly Agree." This scale was used to evaluate key 89 system attributes, including user satisfaction, ease of navigation, system reliability, and efficiency of the GIS and SMS functionalities. Together, these statistical tools provided a comprehensive analysis of user responses, enabling the researchers to draw meaningful conclusions about the system's effectiveness. The results helped identify both the strengths of the system and areas that may require further improvement prior to full-scale implementation.

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